

Review

Unlocking Artificial Intelligence Adoption in Local Governments: Best Practice Lessons from Real-World Implementations

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Abstract: In an era marked by rapid technological progress, the pivotal role of Artificial Intelligence (AI) is increasingly evident across various sectors, including local governments. These governmental bodies are progressively leveraging AI technologies to enhance service delivery to their communities, ranging from simple task automation to more complex engineering endeavours. As more local governments adopt AI, it is imperative to understand the functions, implications, and consequences of these advanced technologies. Despite the growing importance of this domain, a significant gap persists within the scholarly discourse. This study aims to bridge this void by exploring the applications of AI technologies within the context of local government service provision. Through this inquiry, it seeks to generate best practice lessons for local government and smart city initiatives. By conducting a comprehensive review of grey literature, we analysed 262 real-world AI implementations across 170 local governments worldwide. The findings underscore several key points: (a) there has been a consistent upward trajectory in the adoption of AI by local governments over the last decade; (b) local governments from China, the US, and the UK are at the forefront of AI adoption; (c) among local government AI technologies, natural language processing and robotic process automation emerge as the most prevalent ones; (d) local governments primarily deploy AI across 28 distinct services; and (e) information management, back-office work, and transportation and traffic management are leading domains in terms of AI adoption. This study enriches the existing body of knowledge by providing an overview of current AI applications within the sphere of local governance. It offers valuable insights for local government and smart city policymakers and decision-makers considering the adoption, expansion, or refinement of AI technologies in urban service provision. Additionally, it highlights the importance of using these insights to guide the successful integration and optimisation of AI in future local government and smart city projects, ensuring they meet the evolving needs of communities.

Keywords: artificial intelligence; local government; municipality; public services; responsible technology; technology adoption; smart cities



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1. Introduction

Local government, as the closest level of governance body to the community, holds a critical position in ensuring efficient and effective service provision [1,2]. The use of Artificial Intelligence (AI) technologies in various spheres of local government service

delivery has expanded significantly in recent years [3]. This adoption spans a broad spectrum of services, from disseminating information to the public, gathering community feedback, and managing complaints, to tax collection, transportation management, water and sewage management, waste collection and management, and the maintenance of public amenities [4–6]. As local governments increasingly adopt AI technologies [7], thanks to local smart city agendas [8], understanding their functionalities and implications becomes imperative [9].

Understanding the nuances of AI technologies is a crucial process [10,11]. Different AI technologies are designed to address specific tasks or challenges [12,13]. The discernment of which technology is best suited for a particular purpose enables organisations to streamline their workflows, processes, and systems, thereby enhancing efficiency and productivity [14,15]. As AI technologies progressively evolve and permeate across diverse domains, this understanding becomes increasingly critical for local governments navigating complex technological landscapes to realise their aims [16,17]. Specifically, delving into the potential of AI technologies to fulfil service delivery objectives and enhance public welfare is of paramount importance.

There is a growing literature discussing sectoral classification (e.g., health, environment, planning, manufacturing), opportunities and challenges, and the influencing factors of utilising AI technologies in smart cities through literature reviews and policy analyses [18–22]. Despite the rapid growth in AI adoption in smart cities and its potential for positive impact, the scholarly literature offers limited insights into the utilisation of AI technologies in local governments [23]. Moreover, to the best of our knowledge, there exist no studies that comprehensively investigate the practical implementations of AI technologies within local government settings, leaving a significant gap in understanding the real-world application and impact of AI technologies at the local government level. Our study distinguishes itself by taking an empirical approach to investigating real-world examples. The study will provide insight to bridge the gap between the theoretical discussions and practical applications of AI technologies.

As such, this study analyses 262 cases in local governments where AI technologies are utilised, employing a grey literature review approach. These instances are chosen to foster a deep comprehension of the phenomenon of AI adoption in local governments [24,25], and a grey literature review helps mitigate publication bias, fostering a balanced picture of the available evidence [26,27]. Through this approach, this study aims to develop a consolidated understanding of AI utilisation in local government service provision, as well as to generate best practice lessons for similar smart city initiatives.

2. Literature Background

AI is becoming increasingly pervasive and evolving into an umbrella term that encompasses various technological facets [28–30]. This technology set covers a variety of specific segmentations such as computer vision, natural language processing, machine learning [31–34], deep learning, and their generative subsets [35,36], extending its influence across a broad spectrum of domains. This broad scope has rendered the term ‘AI’ inherently vague, making its definition elusive [37]. As AI pervades diverse applied urban domains [38] and industrial spheres [39], it becomes imperative to systematically classify AI technologies [40,41] to effectively understand their functionalities, capabilities, and implications.

Technically, AI was first introduced to the academic literature in 1943 [42]; since then, scientists, researchers, and philosophers have endeavoured to conceptualise and map out AI technologies. Notable contributions include those by [43,44], who provide insights into the AI paradigm. Ref. [43] delineated the history of AI over seven decades from 1930, while [44] mapped the AI history of the last eight decades and offered future predictions based on [45].

In the meantime, the AI knowledge map, developed by [46] and derived by [47], provides an overarching understanding of AI, its subtypes, the problem domains in which

it can be applied, and the capabilities of AI technologies. Figure 1 and Table 1 elucidate the positioning and definition of each technology in the broader AI knowledge realm.

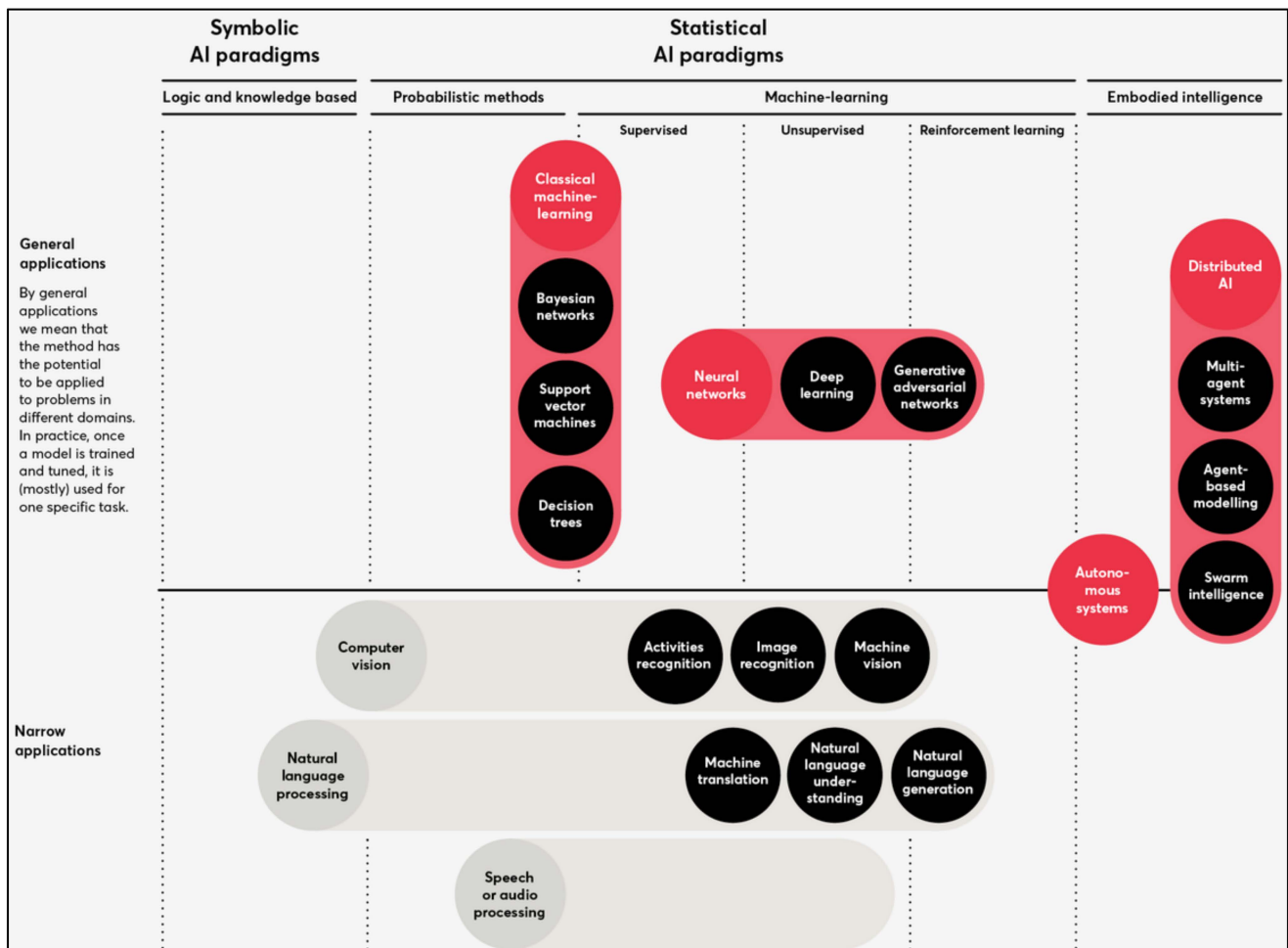


Figure 1. Map of AI knowledge realm [47].

Table 1. Definition of AI technologies.

AI Technology	Definition	Reference
“Inductive Logic Programming (ILP)”	“ILP uses first-order logic to represent data and hypotheses, allowing it to create logical models from real-world data to learn complex relationships”	[48,49]
“Robotic Process Automation (RPA)”	“RPA refers to a technology that enables the automation of business processes using software robots, typically handling repetitive tasks carried out by human workers”	[50,51]
“Expert System (ES)”	“ES simulates the decision-making abilities of a human expert by employing a knowledge-based approach with rules of inference to address problems within a specific domain”	[40,52]

Table 1. Cont.

AI Technology	Definition	Reference
“Decision Network (DN)”	“DN is a type of probabilistic graphical model that can extend such as Bayesian Networks, for example, by incorporating chance nodes, decision nodes, and utility nodes, facilitating effective decision-making in uncertain scenarios”	[53,54]
“Computer Vision (CV)”	“CV enables computers to interpret visual data from the world by using algorithms that recognise patterns, objects, and environments in images and videos, mirroring human visual perception”	[55–57]
“Natural Language Processing (NLP)”	“NLP seeks to empower computers to comprehend, interpret, and respond to human language by analysing the intricacies of language and translating them into computational models”	[58,59]
“Probabilistic Programming (PP)”	“PP is a programming approach designed for dealing with uncertainty in data, where probabilistic models are defined using programming constructs”	[60,61]
“Neural Network (NN)”	“NN, inspired by the human brain for processing data and making decisions, consists of layers of nodes to handle information, including an input layer that receives data, hidden layers for data processing, and an output layer for generating results”	[62–64]
“Affective Computing (AC)”	“AC refers to a digital setting where computational processes are seamlessly integrated into everyday objects and surroundings, becoming an integral aspect of people’s daily lives”	[65,66]
“Autonomous system (AS)”	“AS is an AI system that can operate independently without human intervention”	[67,68]
“Distributed Artificial Intelligence (DAI)”	“DAI represents a category of technologies that fosters collaborative interactions among multiple autonomous intelligent agents, each with distinct capabilities, to solve complex problems”	[69,70]
“Ambient Computing (AmC)”	“AmC refers to a digital setting where computational processes are seamlessly integrated into everyday objects and surroundings, becoming an integral aspect of people’s daily lives”	[65,66]
“Evolutionary Algorithms (EA)”	“EA, inspired by biological evolution, is an optimisation algorithm that employs biomimetic mechanisms to solve tasks beyond the reach of traditional analytical methods within a practical timeframe”	[71,72]

The adoption of AI technologies within the governmental sector is rapidly revolutionising administrative processes, service delivery, and policymaking [7,52]. By leveraging AI technologies, government agencies worldwide are streamlining operations, enhancing citizen services, and making data-driven planning and decisions [29,73,74]. For example, AI-driven chatbots and virtual assistants are increasingly being used to enhance citizen engagement by offering continuous assistance and tailored support, which can help improve accessibility and responsiveness [75,76]. Furthermore, AI technologies address a range of challenges encountered by government entities, including optimising resource allocation, handling large and diverse datasets, mitigating shortages of experts, managing predictable scenarios, and addressing procedural inefficiencies [30,77]. Concurrently, AI is being utilised for citizen inquiries and information exchange, including answering ques-

tions, assisting with document completion and search, sentiment analysis, routing requests, translation services, and drafting documents [78–80].

Accordingly, the exploration of AI adoption and deployment within the public sector has emerged as an increasingly important area of academic interest. Investigations in this area are wide-ranging, including efforts to understand the different ways in which different public sectors are adopting AI, as well as in-depth studies of the opportunities and challenges presented by AI deployment [3,81]. Scholars also examine the proliferation and trends of AI applications across different public sector domains [82,83], factors influencing AI adoption, and employees' perceptions of its implementation. Furthermore, research has explored the synergistic benefits of collaboration between the public and private sectors in AI use [84], the ethical considerations of AI [85], and more.

Simultaneously, governments operate at multiple levels, and each level of government has its own policy priorities, governance structures, and resource constraints [82,86–88]. Academia is interested in understanding how these disparities influence the adoption and implementation processes of AI technologies. Local government, in particular, stands out in this regard. Despite their constrained authority and delineated responsibilities, local governments are increasingly integrating AI into service provision [2]. Nevertheless, research in this area remains nascent and circumscribed.

Recently, Ref. [2] formulated a conceptual framework for responsible AI in local government. Moreover, their subsequent studies have expanded their research to capture city managers' perceptions in Australia and the US regarding the adoption of AI in local government contexts [89], as well as the public perception in Australia and Hong Kong concerning this topic [90]. Similarly, Ref. [91] explored the attitudes and perceptions of officials within the local governments of the Philippines, while [92] analysed the challenges of adopting AI within local government services. Consequently, the findings from these studies indicate the need for further investigation into how AI technologies are being implemented and utilised in local government settings—particularly in the context of smart cities.

3. Research Design

Due to a significant knowledge gap, this study was undertaken to address the research question of 'how AI technologies are being utilised within local government settings'. Data collection occurred in three distinct stages: (a) defining the criteria; (b) searching documents; and (c) filtering documents (Figure 2). The search was conducted in January 2024. To derive the findings, a grey literature review methodology was employed. Grey literature includes information sources not produced through the peer-review processes of scientific journals. It is often defined with specific characteristics unique to various fields [93]. This type of literature contributes to practical applications and helps to extend the range of evidence, filling gaps in academic research [26,27]. For this study, the following method was adopted to acquire grey literature materials.

3.1. Criteria Identification

First, the Boolean search string and inclusion and exclusion criteria were defined. The Boolean string was developed based on the AI knowledge map developed by [46] (Figure 1) and the study area key words. Accordingly, strings were developed separately for each AI technology. For example, ("inductive logic programming") AND ("local government" OR "municipalit*" OR "city*" OR "town" OR "council" OR "borough" OR "Shire"). The used AI technological terms are ("inductive logic programming", "robotic process automation", "expert system", "fuzzy system", "decision networks", "Bayesian program synthesis", "probabilistic programming", "neural networks", "deep learning", "machine learning", "generative adversarial network", "computer vision", "natural language processing", "autonomous system", "distributed artificial intelligence", "decentralised artificial intelligence", "affective computing", "ambient computing", "evolutionary algorithms", "genetic algorithms").

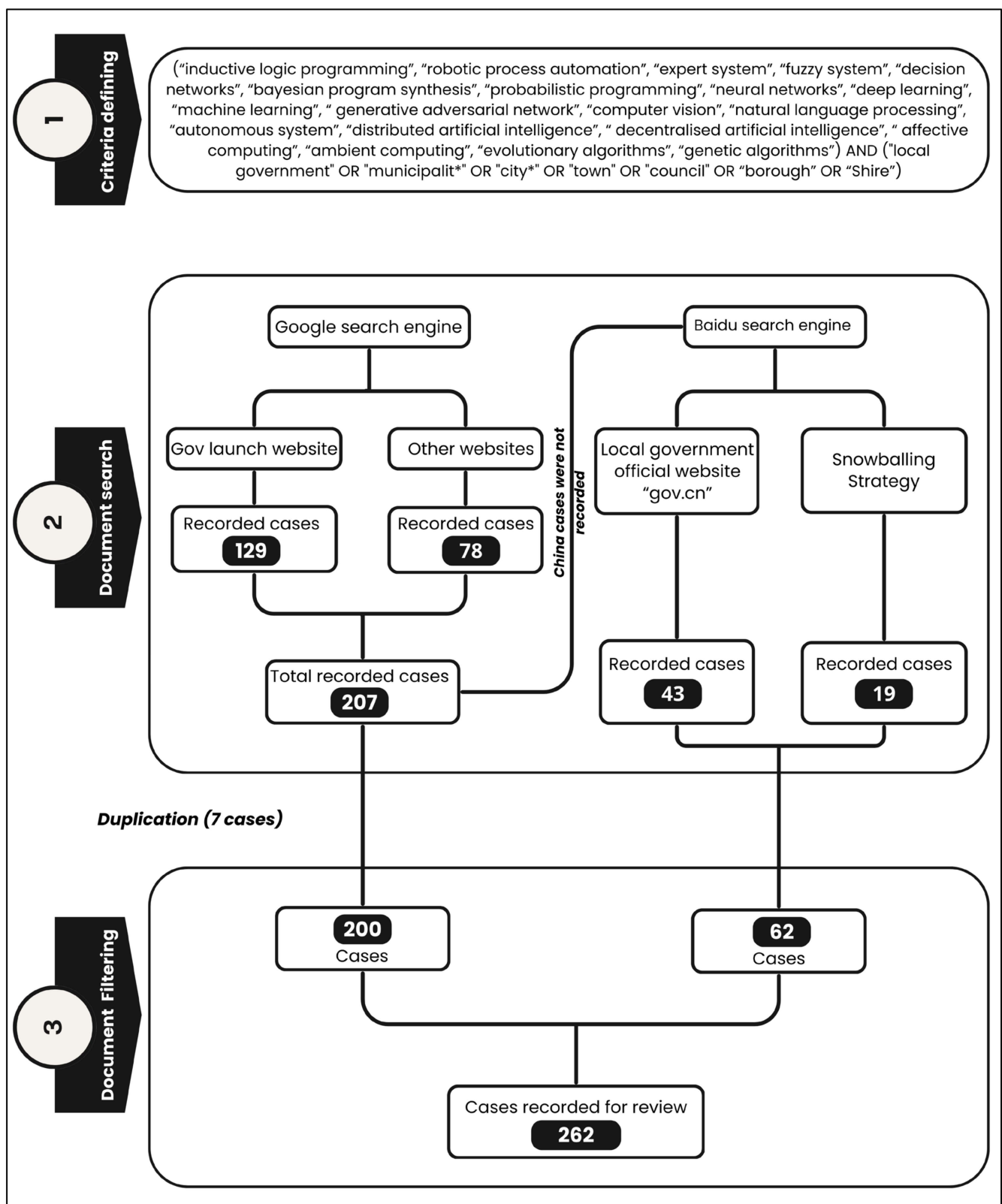


Figure 2. Data collection process.

Next, as shown in Table 2, the inclusion and exclusion criteria were established. The information was gathered from websites, and scholarly journals, book chapters, and conference proceedings have not been taken into consideration to adhere to the research

aim. It is crucial that the cases concentrate on local government service delivery. This study does not consider other public sector agencies and private organisations that operate within the local government area. The timeline was left open in the search criteria for the year of publication, such that the investigation can better understand the evolution in local government AI use and adoption.

Table 2. Inclusion and exclusion criteria.

Inclusion	Exclusion
Published websites, government reports, newsletters, news releases, blogs, technical reports, interviews, etc.	Academic journal articles, books, chapters, conference proceedings
Available online	Unavailable online
Relevant to study aim/question	Not relevant to study aim
In English language	Unavailable in English
Case study: Local government	Case study: National, regional, and other departments or private organisations at local government level
Timeline: open ended	

3.2. Document Search

The principal data collection platform was the Google search engine. Google is the most widely used Internet search engine globally, offering comprehensive access to diverse sources of information with a vast index and advanced search algorithms for decades [94–96]. This is a freely accessible engine, allowing us to gather a wide range of publicly available data without subscription barriers. Accordingly, we employed the above-mentioned search query to search each AI technology separately. Document searches produced thousands of hits in several cases. In these situations, the search was carried out up until the Google alert stated: “In order to show you the most relevant results, we have omitted some entries very similar to the 120 already displayed. If you like, you can repeat the search with the omitted results included”. A useful website employed during the document search stage was “Govlaunch” “<https://govlaunch.com>” (accessed on 10 January 2024). This is a free wiki for innovative local government. Up to January 2024, the website had 8922 instances from all around the world. From this website, 129 local government AI use cases were identified to be included in the analysis.

3.3. Document Filtering

Each case has been systematically recorded within an Excel spreadsheet. For each case, we recorded information against seven categories: (a) AI technology; (b) local government name; (c) country name; (d) comprehensive use description; (e) year of introduction; (f) URL link to the published webpage; and (g) description of services. Subsequently, an additional Google search was conducted to identify the service offered by local governments in the concerned countries. Accordingly, five main service and 28 sub-service categories have been identified (Table 3).

It was found, to our surprise, that none of the local government AI use cases from China had been documented at the ‘country name’ filtering phase, despite the top two countries in AI development and adoption being the US and China [97,98]. Thinking the reason might be Chinese local governments only sharing their information on AI utilisation in Chinese-language and domestic platforms, we conducted a supplementary search task for cases in China via the Baidu search engine to ensure the comprehensiveness of our database. Baidu is the dominant search engine in China and the third most popular search engine used globally (Google and Yahoo! are the top two), giving users access to data and perspectives that might not be available through other search engines [99]. Similar to the other search engines, Baidu offers freely accessible information, making it easier to gather a wide range of publicly available data. Nevertheless, to streamline the search

process, for the case of China, we focused our search on information exclusively from local government’s official website, i.e., information officially announced on the local government website—‘gov.cn’.

Table 3. Local government services.

Main Services	Sub-Services
Administrative services	Information management
	Back-office work
	Community services—complaints
	Community services—interpretation
	Local tax collection
Environmental management	Community feedback
	Waste collection and management
	Maintaining public amenities
	Water and sewerage services
	Local environmental issues
Healthcare and wellbeing services	River management
	Urban forestry
	Public health
	Financial assistance and economic development
	Leisure and recreation
Public safety and law enforcement	Library maintenance
	Burial grounds and electric crematorium
	Pest control services
	Public safety and security
	Meteorological services
Transportation and urban planning	Transportation and traffic management
	Permits granting and licensing
	Resident registry
	Housing services
	Town planning
	Building regulations
	Local road maintenance
	Planning application processing

In addition, a snowballing strategy was adopted to achieve a comprehensive coverage of cases [100]. Finally, a total of 62 cases in China were identified. Each case has been manually reviewed to ensure the thoroughness and precision of the materials, focusing on several critical aspects: the reliability of the source of information, contextual relevance, and alignment with the inclusion criteria. This meticulous process guarantees that only the most pertinent and rigorously evaluated data are included, maintaining the integrity and validity of our study. Lastly, duplicate entries were removed. This leaved us with 262 AI use cases from 170 local governments—forming a local government AI use case dataset (see Appendix A).

4. Analysis and Results

4.1. General Observations

The local government AI use case dataset developed through the grey literature review in January 2024 encompasses 262 cases, spanning 170 local governments. Despite the existence of numerous AI technologies (Figure 1), the records indicate that only eight technologies are actively utilised within local government settings. Among these, NLP emerges as the most prevalent, accounting for 108 cases, followed by RPA with 58 cases, NN with 47 cases, CV with 36 cases, and AS with 10 cases. Conversely, AC, AmC and ILP

exhibit the lowest levels of utilisation, each with only one documented use case, as shown in Table 4.

Table 4. AI technologies and local government use cases.

AI Technology	Use Case Number
Natural language processing (NLP)	108
Robotic process automation (RAP)	58
Neural network (NN)	47
Computer vision (CV)	36
Autonomous system (AS)	10
Affective computing (AC)	1
Ambient computing (AmC)	1
Inductive logic programming (ILP)	1
Total	262

Table 5 presents the local governments with three or more documented use cases of AI technologies. Notably, the top seven positions are occupied by local governments in China, with Changsha leading with eleven cases, followed by Hangzhou with nine cases, and Shenzhen with seven cases. Additionally, 22 local governments are recorded with two cases, while 128 local governments have one documented case each.

Table 5. Local governments with more than two AI use-cases.

Local Government	Country	Use Case Number
Changsha	China	11
Hangzhou	China	9
Shenzhen	China	7
Chongqing	China	6
Shanghai	China	6
Beijing	China	5
Guangzhou	China	4
Kawasaki	Japan	4
Los Angeles	US	4
North Tyneside Council	UK	4
Adelaide	Australia	3
Buenos Aires	Argentina	3
Chengdu	China	3
Derby City Council	UK	3
Helsingborg Municipality	Sweden	3
Johns Creek	US	3
Ronneby	Sweden	3
Singapore	Singapore	3
Telford and Wrekin Council	UK	3
Williamsburg	US	3

Analysing the distribution of AI utilisation cases in local governments by country, it is observed that more than half of the cases originate from China, the US, and the UK, totalling 160 cases. Specifically, China leads with 62 cases, followed by the US with 53 cases, and the UK is in third place with 45 cases. Australia, Sweden, and Canada also exhibit a notable presence with 22, 13, and 12 cases, respectively (Figure 3). Furthermore, the utilisation of AI technologies varies across countries. In the US, NLP emerges as the predominant AI technology, with 33 cases recorded in local governments. Conversely, the UK exhibits a balanced adoption of RPA and NLP, totalling 17 cases. Notably, China predominantly utilises NNs, with 25 cases, and CV, with 16 cases, as the primary AI technologies employed within its local governments (Figure 4).

The earliest use case was from the year 2004. However, not many use cases were documented between 2004–2010 and 2010–2014. But a steady increase in recorded cases

is observed from 2014 onwards, with a doubling of documented cases between 2017 and 2018, signifying the onset of an exponential trend in growth. Despite a slight decline in 2019 with 22 cases, there was a significant surge in 2020, with 58 cases recorded. The peak in recorded cases occurred in 2021, totalling 68 cases (Figure 5).

In terms of technology adoption trends, NLP has demonstrated consistent operational usage in local government settings from 2004 onwards, maintaining its presence until 2024. Conversely, the adoption of other technologies exhibits fluctuating growth patterns. According to recorded cases, RPA was introduced in 2015, CV and ASs in 2016, and NNs in 2017. Additionally, AC was introduced in 2016 and ILP and AmC in 2019 (Figure 6). It is interesting that the number of cases is dropping off across the board from 2021 to 2024. One possible reason behind this could be a lag behind the publication of such AI initiatives from local government.

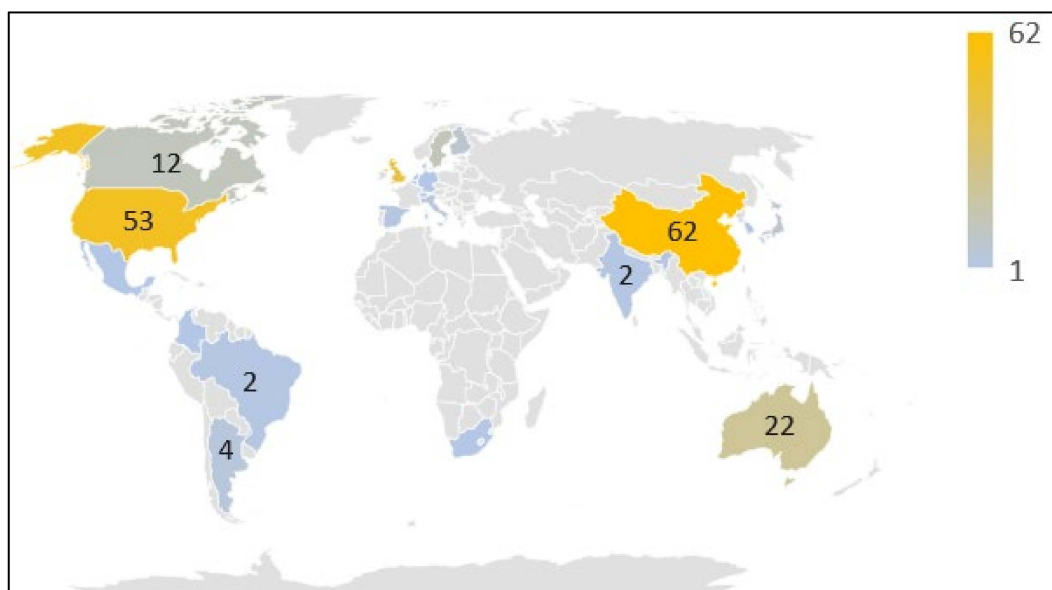


Figure 3. Use cases by country.

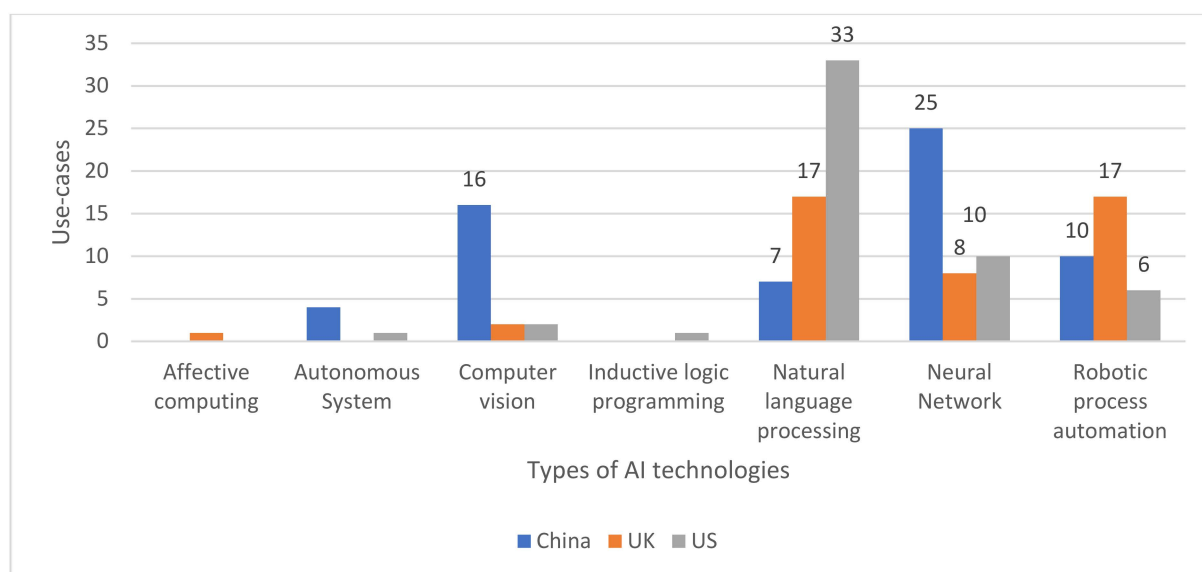


Figure 4. AI technology by US, UK, and China.

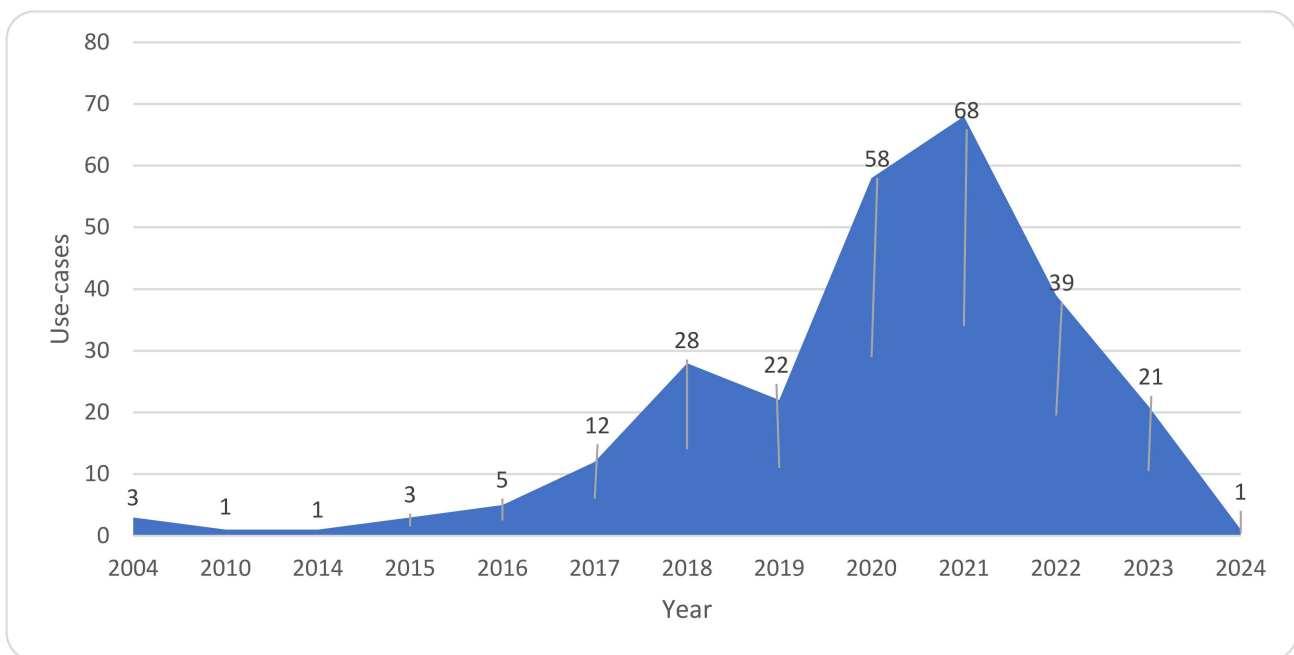


Figure 5. Use cases by year (all AI technologies).

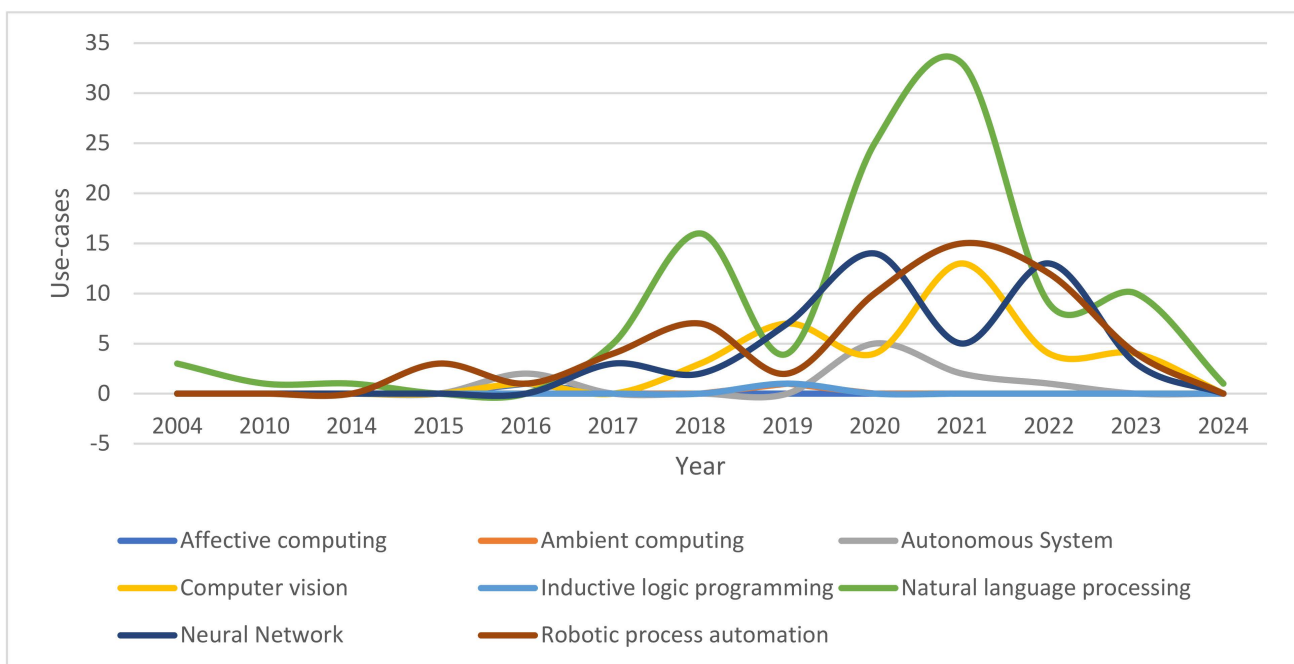


Figure 6. Use cases by AI technology and year.

As previously mentioned, a total of 28 sub-service categories have been documented under five main services within the dataset. Notably, information management emerges as the most prevalent area with 49 recorded cases, followed by back-office work with 33 cases, transportation and traffic management with 27 cases, and public health with 25 cases. It is noteworthy that local governments are actively adopting AI technology across a wide array of domains, reflecting a concerted effort towards enhancing service delivery effectiveness and efficiency (Table 6).

Figure 7 illustrates the distribution of years and services within the dataset. It is evident that certain years are associated with specific services. For instance, in 2018, four main branches are connected, namely back-office work, information management, permits granting and licensing, and transportation and traffic management. Similarly, in 2020, connections are observed primarily towards three main services, namely, back-office work, information management, and public health. Furthermore, in 2021, links are established between back-office work, information management, public health, and waste collection and management. Lastly, in 2022, connections are noted between back-office work, information management, transportation and traffic management, and waste collection and management.

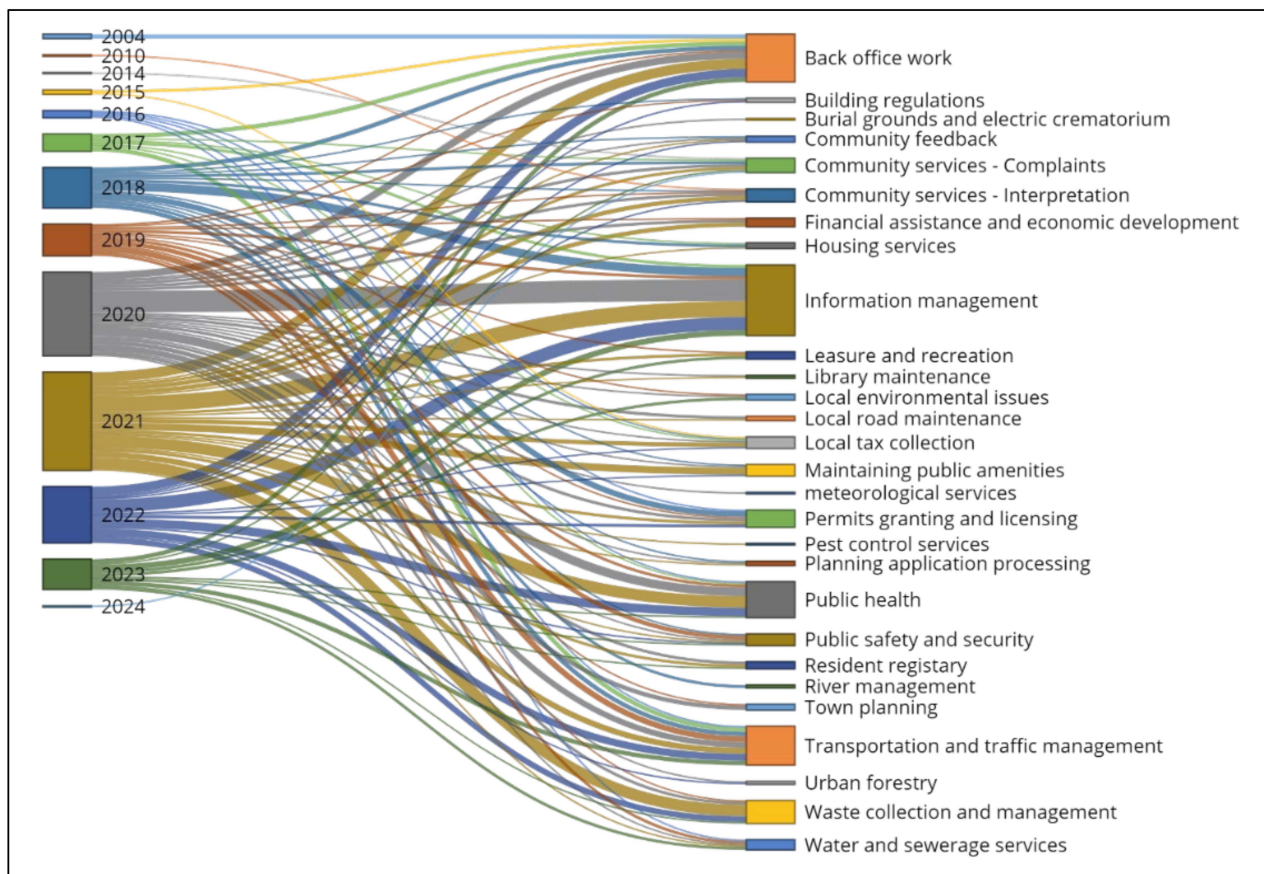


Figure 7. Local government services supported with AI by year.

Table 6. Service by use cases.

Service	Use Case Number
Information management	49
Back-office work	33
Transportation and traffic management	27
Public health	25
Waste collection and management	16
Permits granting and licensing	12

Table 6. Cont.

Service	Use Case Number
Community services—complaints	10
Community services—interpretation	9
Local tax collection	8
Maintaining public amenities	8
Public safety and security	8
Water and sewerage services	7
Financial assistance and economic development	6
Leisure and recreation	5
Resident registry	5
Community feedback	4
Housing services	4
Local environmental issues	4
Town planning	4
Building regulations	3
Local road maintenance	3
Planning application processing	3
Library maintenance	2
River management	2
Urban forestry	2
Burial grounds and electric crematorium	1
Meteorological services	1
Pest control services	1

4.2. AI Technology and Service Distribution

4.2.1. Natural Language Processing in Local Governments

The results of the analysis indicate that NLP has been connected with nearly 18 services in local government. Among them, information management, back-office work, posting complaints, interpretation, and public health are the highly provided services. NLP is the technology that has been used in chatbots to understand natural human language communication (Figure 8).

These capabilities of NLP provided local government with a more effective and efficient way of doing the following:

- Removing language barriers—Phoenix Council, US, utilises the Amazon Web Services (AWS) Lex chatbot to create a conversation interface in both English and Spanish [101];
- Freeing up human time from performing repetitive boring tasks—Lewes and Eastbourne Council in the UK employ ELLIS, which covers over 1000 council topics and which was trained on 12,000 resident questions. It has already enabled the relocation of 5 full-time contact agents away from live chat to focus on more complicated tasks [102];
- Connecting residents to city council services 24 h a day—the Public Relations Office within the Municipality of Grosseto in Italy implemented digital functions to enhance communication between residents and the administration. A virtual assistant is available 24/7 to guide residents through online procedures and assist with problem solving [103];
- Enhancing wide-scale customer experience—the municipalities of Kortrijk, Tournai, and Roubaix collaborated to create the free Tripster chat tourism service, an overarching approach to promoting cross-border tourism and making it more accessible to everyone [104].

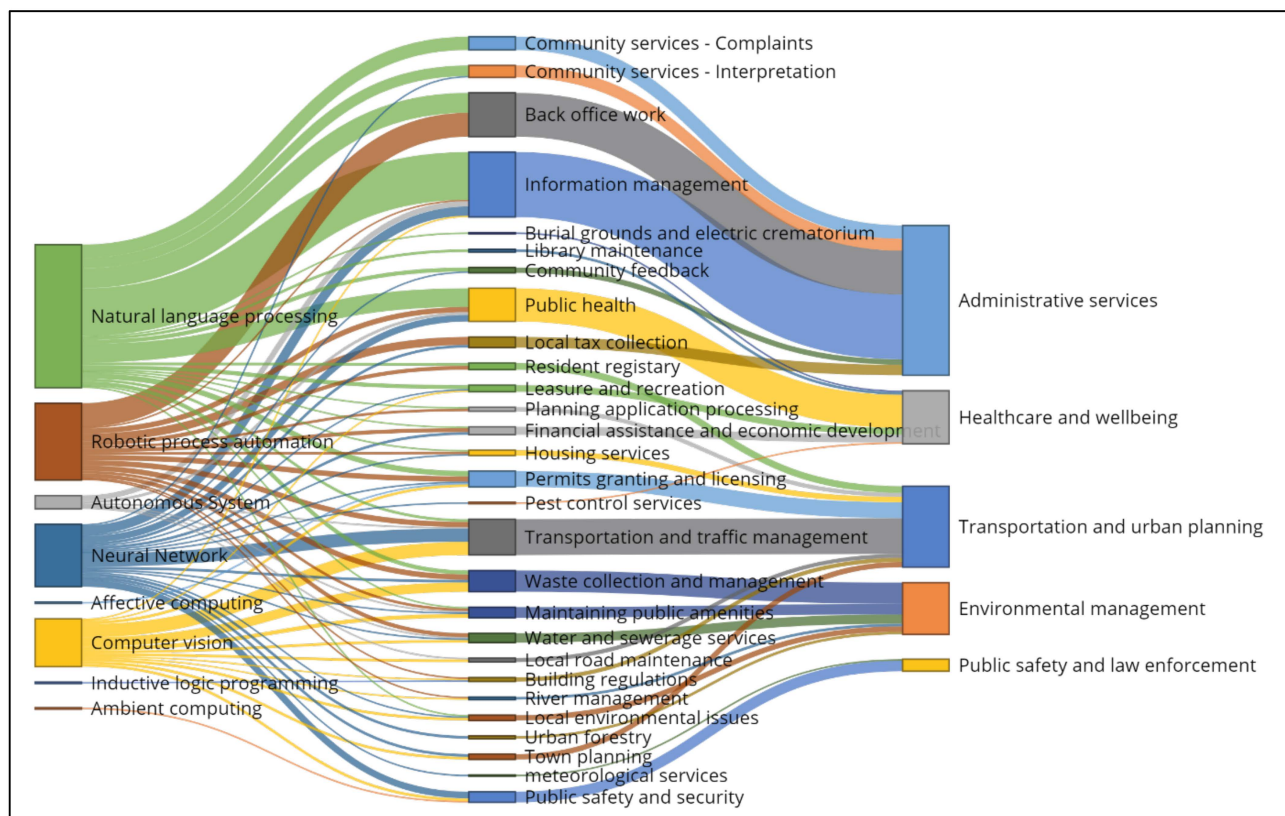


Figure 8. AI technology and AI-supported local government services.

Additionally, NLP has played a key role during the COVID-19 pandemic. The Kolkata Municipality in India used an innovative chatbot tool to streamline the vaccination process as it fought COVID-19. Within ten weeks of the chatbot-embedded platform launch in mid-May 2021, it attracted more than 250,000 unique users and booked more than 75,000 vaccination appointments directly through the platform. Initially, the platform was only connected to three vaccination centres. However, it was rapidly expanded to include approximately 100 vaccination sites across the city, which significantly improved the accessibility and efficiency of the vaccination campaign [105]. Furthermore, some municipalities have leveraged NLP for short-term purposes, such as during elections. For instance, the Hamilton Council in Canada introduced two innovative online tools—a voice query directory and a virtual assistant—aimed at enabling residents to effortlessly locate and access information about the municipal election held on 24 October 2022 [106].

4.2.2. Neural Networks in Local Governments

NNs are integrated with 19 services and predominantly linked to transportation and traffic management, public safety and security, information management and public health. However, NNs are recognised as a computationally expensive technology due to their demand for significant processing power and time. Consequently, they have been utilised for complex service provision such as transportation and traffic management and public safety and security.

Several municipalities utilise NNs for different types of transportation and traffic management, including the following:

- Mapping ideal locations for electric vehicle charging points—implemented by Irving Municipality in the US [107];
- Junction improvements—undertaken by Lancashire County Council in the UK [108];
- Determining the safest route—implemented by Los Angeles City Council in the US [109];

- Assisting citizens in emergency situations such as bridge collapse—utilised by Atlanta City Council, US [110];
- Analysing traffic patterns of different mode of transportation—implemented by Kansas City Council in the US [111];
- Navigating parking systems—utilised by Hangzhou Municipality [112], among others.
- Moreover, NNs have also been utilised for safety and security, including the following:
- Predicting crime locations—Chicago Municipality adopted a model to predict when and where violent crimes are likely to occur. The former mayor Rahm Emanuel announced in early 2018 that gun violence was down 25% compared to the previous year [113];
- Predicting child abuse—implemented by Hackney Council in the UK [114];
- Safeguarding against cybersecurity issues—utilised by Gilbert Town Council in the US [115];
- Identifying and addressing anti-social behaviours—undertaken by Sunderland City Council in the UK [116].

4.2.3. Robotic Process Automation in Local Governments

RPA relates to 13 services, with 51 recorded cases of its usage. Among these cases, the majority were recorded for back-office work. The back-office is often referred to as the engine room of an organisation [117], where much of the work performed determines the overall success of operations [118,119]. Its tasks encompass procurement, finance and accounting, human resource management, payroll, work reporting, and more [117,120,121].

Local government back-office employees often find themselves engaged in repetitive tasks for many hours each day, which can lead to an acceleration of errors and a slowdown in progress [5,122]. In addition, inefficient legacy systems contribute to the accumulation of numerous pre-approval documents on desks for extended periods, resulting in suboptimal service delivery. The presence of outdated and inefficient administrative processes not only hampers the speed at which local governments can respond to the needs of residents but also affects the overall quality of services provided [123,124].

RPA is recommended technology for back-office work, and this finding is further justified in the local government sector as well [125]. It offers a non-invasive and cost-effective solution [126,127], which is particularly important for local councils operating in high-paced work environments with limited budgets [128,129].

In local government back-office work, RPA is utilised for various tasks, including the following:

- Payslip account management, including a council's payslip archiving system—implemented by Surrey County Council in the UK [130];
- Management of financial assistance processing—undertaken by Strängnäs Municipality in Sweden [131];
- Validation of Blue Badge applications and invoice processing—managed by Cumbria County Council, UK [132];
- Financial auditing and risk management mitigation—Liverpool City Council, Australia [133];
- Mileage calculations and value added tax (VAT) calculations—handled by Gloucestershire County Council [134].

Tax calculation [135], water and sewerage service [136], waste collection [137] and river management [138] are among other main services provided by RPA in local government.

4.2.4. Computer Vision in Local Governments

According to the findings, CV has been employed across 13 services in local government. Among these, CV is predominantly used for transportation and traffic management, as well as waste collection and management services. Some local governments have also utilised CV for transportation management, gradually extending its application to public safety. For instance, Seoul Municipality fixed CCTV on every street, which transmit data to the Transportation Operation and Information Service (TOPIS). The TOPIS website utilises this data for real-time traffic monitoring. Additionally, instances of illegal vehicle driving or parking are detected, leading to automatic fines. In cases of accidents or road construction, detour routes are suggested, and accident notices are promptly sent to connected police and hospitals (Bandopadhyay, 2019). Moreover, there are local government AI use cases in remote sensing to count pools, assess rooftop solar panels, for electrical infrastructure asset management, leak detection in water management, and in the security/surveillance space.

Some local governments have installed fixed cameras to municipal garbage tracks to identify the following:

- Roadside assets maintenance—Brimbank City Council, Australia [139];
- Pothole detection—Helsingborg Municipality, Sweden [140];
- Identification of blighted areas—Tuscaloosa Municipality, the US [141].

4.2.5. Other AI Technologies in Local Governments

Autonomous systems are used for five types of services, primarily focusing on information management in local government. For instance, Ogaki City Council in Japan uses robots to guide people to the appropriate information window or assist them in filling out government forms [142]. Additionally, affective computing, ambient computing [143,144], and ILP are used for each service of local government. Affective computing is used for permit granting and licensing by London Borough Council in the UK [145]. ILP is employed to codify building regulations in California municipalities [146]. In addition, ambient computing is utilised for public safety and security in Australia [147].

5. Findings and Discussion

5.1. *Why Have NLP and RPA Gained Popularity in Local Governments, and How Can These Technologies Address Specific Challenges?*

NLP and RPA have gained popularity in local governments due to their profound impact on efficiency and communication. The adoption of NLP significantly enhances the quality of interaction between governments and citizens in administrative processes/governmental affairs procedures [148,149]. The common application is the NLP-driven chatbots and virtual assistants that are operational round the clock, efficiently addressing frequently asked questions (FAQs) and navigating users through intricate administrative procedures [150,151].

These automations streamline information access and service requests and contribute to boosting citizen satisfaction with the government's service experiences [152–154]. Furthermore, NLP's capability to sift through and analyse copious amounts of textual data accrued in administrative processes is invaluable [150,155]. This process facilitates the extraction of pertinent insights and the discernment of trends, and informs governmental policy decisions [89,149,156]. An exemplar of its application is sentiment analysis, which can efficiently evaluate public opinion on diverse issues, thus empowering governments to tailor their responses to citizen concerns more efficiently, while enhancing the government's response level [157]. This refined approach to public engagement and data analysis underscores NLP's pivotal role in modernising and optimising government–citizen interactions [148,151].

RPA has become a crucial instrument in streamlining the routine tasks prevalent in local government operations, particularly in the context of implementing digital transformation initiatives within smart city strategies [158]. RPA's ability to automate paperwork and repetitive administrative functions significantly alleviates staff workloads [159,160]. This automation translates into expedited processing for various administrative pro-

cesses/governmental affairs procedures, such as license renewals and application processing, enhancing overall service delivery [161]. A key benefit of RPA is its potential for cost reduction, a critical factor for local governments operating within stringent budgetary constraints [162].

Additionally, the precision of automated processes reduces the likelihood of errors, ensuring a heightened accuracy in data management and regulatory compliance—a vital consideration in government operations where inaccuracies can lead to substantial repercussions [158,159]. Thus, the adoption of NLP and RPA technologies offers local governments a pathway to not only bolster operational efficiency and service quality but also promote government decisions to adapt more responsively to the dynamic needs and expectations of their citizenry, e.g., unbiased decisions, new forms of democratic participation, inclusion of users, and improved working conditions for employees [160,163,164].

5.2. Which Service Areas Are Most Affected by AI Technology in Local Governments, and How Does AI Improve the Efficiency in These Service Areas?

AI technology is revolutionising various service areas within local government, with significant impacts observed in the areas of administrative services, healthcare and well-being, transportation and urban planning, environmental management, and public safety and law enforcement:

Firstly, AI facilitates the automation of routine administrative tasks in local government operations, such as information management, back-office work (e.g., invoice and application processing), community services for addressing complaints and interpretation, local tax collection, and community feedback. Information management involves organising, storing, and retrieving data efficiently. AI assists in managing vast amounts of data by automating data categorisation, ensuring accurate record keeping, and enabling quick access to necessary information. Similarly, AI systems handle tasks such as invoice and application processing with greater speed and accuracy than manual methods. This process reduces manual data entry errors and speeds up processing times for various administrative tasks [165]. For example, at Cumbria County Council, the invoice-processing system was transformed from a manual process, involving scanning invoices, to a new web form that generates emails with attached documents for audit. This change has eliminated the need for printing and scanning, saving over 1000 kg of paper in three months, and freed up staff time previously spent on attaching stickers to printouts, allowing them to focus on complex transactions [132]. NLP algorithms enable chatbots and virtual assistants to handle citizen queries effectively, reducing response times [7,166]. AI-based workflow automation platforms streamline administrative processes by routing tasks, assigning priorities, and automating notifications and approvals [167–169]. These systems optimise task management, reduce bottlenecks, and ensure smoother coordination among local government officials, thereby leading to heightened productivity and efficiency [17].

Secondly, AI transforms transportation and urban planning by optimising traffic flow, enhancing public transit, implementing smart parking solutions, refining infrastructure planning, integrating micro-mobility options, promoting transportation equity, and dealing with local road maintenance. Smart traffic management systems use AI to analyse real-time traffic patterns and accordingly adjust signal timings to improve flow, reduce congestion, and modify speed. For example, municipalities like Seoul improve public transportation systems by optimising routes and schedules based on passenger demand and traffic conditions [170]. Additionally, AI-powered apps provide real-time updates to passengers, enhancing their travel experience. In urban planning, AI is used to simulate scenarios for infrastructure development, optimising city layouts for improved mobility, dealing with permits granting and licensing, codifying planning rules and regulations, and processing planning applications. For instance, the Rotterdam Municipality uses a 3D simulator to assess the vulnerability of buildings during flooding. By knowing the exact location and elevation of doors in a 3D model, urban planners and engineers can assess how susceptible a building is to flood. If the sills are low, the building is at a higher risk during heavy

rains or rising water levels [171]. These AI-driven solutions improve efficiency, reduce congestion, minimise emissions, and foster inclusive urban development, creating more sustainable and resilient local government functions [172,173].

Thirdly, effective environmental management is crucial for maintaining public health, safety, and sustainable urban living. As populations grow and consumption patterns evolve, the volume and complexity of waste increases, presenting significant challenges for municipalities [174]. Traditional environmental management methods often struggle to keep pace with these demands, leading to inefficiencies, increased pollution, and resource wastage [175]. In local government, AI enhances environmental management by tracking pollution sources, optimising waste management and water and sewage management, enhancing energy efficiency, conserving natural resources, aiding climate adaptation, and supporting emergency responses. For example, Hangzhou Municipality, in China, employs AI and big data to create an intelligent waste management system that improves the accuracy and efficiency of domestic waste classification. The system integrates digital tools for training and real-time monitoring, resulting in a significant improvement in waste-sorting accuracy. Also, it greatly reduces the secondary pollution caused by landfill, incineration, and other disposal processes, and realises the coordination and efficiency of pollution reduction and recycling [176]. Moreover, AI aids in conserving natural resources by optimising their use and management. The Chongqing municipality of China integrates technologies such as AI, BIM, 3D GIS, and the IoT, and big data is used to achieve the comprehensive digitalisation and intelligent monitoring of Guangyang Island's ecosystems, including mountains, rivers, forests, farmlands, lakes, and grasslands. This creates a global sensing and accurate mapping ecological monitoring system. The system can automatically identify and alert authorities to tree vegetation insect damage, generate and dispatch work orders, and supervise the entire process [177]. Meantime, AI supports emergency response efforts by providing real-time data and predictive analytics. During natural disasters or environmental emergencies, AI systems can offer valuable insights into the severity and trajectory of the event, enabling local authorities to allocate resources efficiently and coordinate evacuation efforts more effectively. Ku-ring-gai Council, Australia, undertook a project, which uses ambient computing technology to stimulate extreme fire weather, to increase engagement in their Climate Wise Communities (CWC) workshop program [147]. By leveraging AI across these fronts, local governments enact proactive measures, reduce environmental impact, and promote sustainability, safeguarding communities and fostering resilience for future generations [178].

Fourthly, AI revolutionises the healthcare and wellbeing service of local governments by enhancing public health initiatives, optimising financial assistance programs, supporting economic development, improving leisure and recreational services, ensuring efficient library maintenance, managing burial grounds and electric crematoria, and implementing effective pest control services. AI-powered chatbots and virtual assistants interact with citizens, providing information on healthcare services, scheduling appointments, and offering initial triage for medical inquiries [122,179]. This reduces the burden on staff, improves accessibility to services, and ensures timely assistance for citizens. The COVID-19 pandemic has prompted local government entities to significantly utilise AI technologies to bolster health and safety measures [180]. For example, the Boston Mayor's Office of Food Access developed a food delivery chatbot during COVID-19 to provide a contactless food delivery service addressing hunger issues. This SMS-based service requires no internet connection, making it more accessible than web or app-based chatbots. It was available in the city's eight most spoken languages [181]. Meantime, financial assistance programs are also optimised through AI, streamlining the allocation of resources and reducing economic disparities. National legislation enables residents to request 60-day freezes on debt enforcement. Norwich City Council, England, employed RPA across 25 services to automate responses to Debt Respite Scheme requests. Approved requests trigger debt pauses in six departments, with real-time updates reflecting debtor interactions with the council [182]. Furthermore, AI optimises leisure and recreational services by facilitating

the efficient maintenance of parks, sports facilities, and libraries. Through continuous monitoring of usage patterns and predictive analytics, local governments can anticipate maintenance requirements in advance. In Stirling municipality, signs with QR codes for a chatbot were installed at 42 locations including libraries, basketball courts, and community centres. The chatbot asks questions about visitor experiences, facility conditions, and areas of improvement by text message. The chatbot can also answer questions about city facilities and amenities [183]. This proactive approach ensures that public spaces are consistently well-maintained, fostering safe and engaging environments that promote both mental and physical health among community members [179,184].

Finally, AI-powered chatbots and virtual assistants engage with communities, offering crime prevention tips, safety information, and aid in reporting incidents, fostering trust, transparency, and collaboration between local government and citizens for more effective emergency services [185]. AI systems prioritise emergency calls, assess incident severity, and recommend response strategies, enhancing dispatch efficiency, reducing response times, and optimising resource allocation during emergencies [186,187]. For example, Seoul's smart city initiatives focused on deploying computer vision-powered surveillance and public safety systems. This included the widespread use of surveillance cameras and real-time monitoring, resulting in a notable decrease in crime rates and fostering a greater sense of safety among residents. By investing in these technologies, Seoul not only enhanced public safety but also directly contributed to an improved quality of life through a safer urban environment [188].

In sum, local governments can significantly enhance operational efficiency by leveraging AI to automate tasks, analyse data, predict trends, enhance citizen engagement, and optimise resource allocation. This enhanced efficiency facilitates more effective public service delivery, enabling the tackling of complex challenges with greater precision and agility. The integration of AI technologies into the operational frameworks of local governments not only streamlines administrative processes but also enriches the decision-making ecosystem with data-driven insights. This technological empowerment facilitates a more responsive, transparent, and inclusive governance model and contributes to fostering a more connected and satisfied community.

5.3. Why Do Public Safety and Law Enforcement Get Less Attention for Local Government AI Applications?

Local governments often prioritise AI applications that directly address immediate challenges or offer clear efficiency gains in service delivery [3,29]. The results indicate that AI initiatives in areas such as administrative services, transportation, and urban planning may receive more attention than public safety. Meanwhile, public safety issues are often multifaceted and sensitive, involving considerations of law enforcement, emergency responses, community relations, and privacy rights [189,190].

Implementing AI solutions in this domain requires careful planning, stakeholder engagement, and consideration of ethical and legal implications, which can pose challenges for local government AI projects [191]. Local governments often face resource constraints in terms of both funding and expertise when it comes to developing and implementing AI solutions for enhancing public safety [3,73]. Consequently, they prioritise their investments based on immediate needs and may allocate resources to other services. For instance, Seoul Municipality initially installed CCTV cameras for real-time traffic monitoring. However, during emergencies, they repurpose these cameras to enhance city safety, demonstrating a flexible and adaptive approach to resource utilisation.

In the newsletter of *Forbes*, Dan Hoffman, the city manager of the City of Winchester, stated that "...in time though, I see AI having an impact as big, if not bigger, in public safety. However, those will take longer to develop and trust for obvious reasons. Adoption in the first responder community will also take time as those systems are traditionally more expensive and require more standards and training. However, when the time comes, there will be huge leaps in AI tools that help prevent fires and medical emergencies. We're

already seeing AI tools employed by groups like the National Centre for Missing and Exploited Children making great strides in fighting child sex abuse. So, we're seeing larger national organisations use AI for public safety; it's just a matter of time before it is more common at the local level" (Schmelzer, 2020) [192].

Furthermore, addressing the concerns and challenges associated with implementing AI in public safety necessitates a long-term strategy incorporating input from a broad array of stakeholders and establishing ethical guidelines. This strategy is required to guarantee that AI technologies are deployed in a manner that is transparent, responsible, and effective, especially within community settings. As [193] suggested, AI-driven applications in public safety require a deliberate and considered framework that balances technological innovation with ethical considerations, safeguarding individuals' and communities' rights and wellbeing, while enhancing public safety measures.

5.4. Why Do Different Local Governments Use Different AI Systems?

The findings reveal a diverse landscape where different local governments employ various AI systems for service provision. This diversity stems from the unique needs and priorities of each locality, shaped by factors like population size, geographic location, economic activity, and social demographics [194,195]. For instance, a city prone to natural disasters may prioritise AI systems for disaster prediction and prevention, while an urban area may focus on urban issues such as traffic optimisation. The selection of specific AI systems is influenced by a myriad of factors (Duan et al., 2019) [196], including budgetary constraints, the availability of technological infrastructure, and human capital [197–199]. Local governments with robust resource allocation mechanisms tend to leverage AI systems more effectively and efficiently [200].

Similarly, local governments design their AI systems to align with specific policy goals and objectives [5]. Urban areas might prioritise AI for traffic management, public safety, and smart city initiatives to enhance urban living conditions and sustainability. In contrast, rural areas may focus on agricultural monitoring, resource management, and improving healthcare accessibility [20]. These policy-driven objectives ensure that AI technologies are strategically implemented to meet the distinct needs and challenges of each locality.

Furthermore, each local government grapples with its unique set of challenges, spanning from crime rates and environmental concerns to transportation congestion [201,202]. In response, AI systems are tailored to address these specific challenges, resulting in varied deployment strategies based on local needs [203]. As primary service providers to the community, local governments must accommodate the preferences of a diverse range of stakeholders, including community members themselves [204]. However, aligning these preferences can be complex, as public officials, community leaders, and citizen stakeholders may harbor differing priorities and opinions regarding AI adoption [205]. Moreover, local governance structures and political dynamics significantly influence the selection and implementation of AI systems [206].

Overall, the adoption of various AI systems across different local governments is driven by a multitude of factors, including distinct needs, resources, regulations, and stakeholder preferences. This diversity necessitates a thorough evaluation by each local government to select AI solutions that best align with their specific challenges and objectives. It is imperative for AI deployment in local governments to not only address the unique needs of communities but also to comply with local regulatory frameworks and to accommodate the expectations of diverse stakeholders.

Strategic partnerships play a vital role in this process by providing critical resources, expertise, and technology that local governments might not possess independently [2]. Additionally, government grants and international programs targeting specific local government policy areas, such as smart cities or digital transformation, further guide AI adoption, ensuring alignment with broader policy objectives. Tailoring AI deployment in this manner can enhance the capacity of local governments for service delivery, informed decision making, and overall operational efficiency, particularly within specific community contexts.

5.5. What Are the Key Challenges, Future Impacts, and Trends?

The adoption of AI technology in local government holds promise for enhanced efficiency and service delivery, but also poses significant challenges that demand deliberate, careful consideration and strategic planning. One major hurdle lies in integrating AI technologies with existing legacy systems [29,165,207], as these systems may not be designed for compatibility with AI. This integration process often requires extensive overhauls or replacements and entails seamless functionality and communication between new and established systems [29,208]. Additionally, since AI-driven applications rely heavily on data, safeguarding the privacy and security of sensitive information managed by governments becomes a critical concern [154,207,209].

This challenge includes considerations not only of the technical aspects of data security but also of ethical dimensions and the preservation of public trust [210–212]. Moreover, the potential of AI algorithms to unintentionally reflect and amplify biases present in their training data necessitates vigilant oversight of the ethical repercussions of AI-driven decision making [213,214]. This oversight is imperative to ensure the equitable and non-discriminatory application and practice of these technologies [215,216]. Another pivotal challenge is navigating the evolving regulatory landscape, particularly concerning data usage and privacy. Governmental entities are compelled to ensure adherence to extant laws, while remaining adaptable to a dynamic regulatory context [207,217,218]. As regulations governing data privacy and use continue to evolve, local governments must stay informed of these changes and adapt their practices accordingly. This adaptation is critical to ensure compliance with legal standards and uphold ethical principles in the implementation of AI technologies.

Furthermore, the financial consideration of implementing AI solutions may pose a significant barrier, especially given the budgetary constraints typical of local governments [84,90]. In this context, it is imperative for local governments to meticulously evaluate the balance between financial outlays and the potential long-term advantages of AI implementation. This evaluation must prioritise ensuring that the realised benefits align not only with the initial promises of AI adoption and utilisation but also substantively contribute to enhancing the wellbeing of citizens [219]. It is critical that these advancements are achieved without inadvertently compromising the welfare of any community segments, thereby maintaining a holistic approach to public service enhancement through AI integration [220–222].

Lastly, it is important to note local government cybersecurity [223] and the role of fake/synthetic data/poisoned data and adversarial attacks, and how these may affect services provided through AI-enabled local government systems are among the factors to be carefully considered.

As highlighted previously, issues such as data privacy concerns, limited technical expertise, and financial constraints pose significant barriers to the seamless adoption of AI-driven solutions. Addressing these challenges requires effective strategies to involve the following:

- *Policy framework:* Establishing a robust policy framework is essential for guiding the ethical deployment and responsible use of AI technologies within local governments [156]. By defining policies that uphold responsible ethical principles of transparency, fairness, equity, and inclusivity, etc, local governments can ensure that AI systems enhance public trust and serve community interests effectively [215].
- *Capacity building:* The deployment of AI technologies requires a workforce with specialised skills, not only in terms of recruiting new talent but also in training existing employees to operate and manage these advanced systems competently [84,224]. This need for skill development may represent a time-intensive and financially demanding undertaking, adding to the complexity of adopting AI in local government. It underscores the importance of investing in human capital development alongside technological infrastructure to ensure the successful deployment and sustainable functioning of AI solutions within local government contexts.

- *Financial support:* Securing adequate financial support is pivotal for the successful implementation and sustainability of AI projects within local governments. Financial resources are needed to invest in AI infrastructure, software licenses, data storage solutions, and ongoing maintenance costs [84,90]. Local governments can explore various funding mechanisms, including public–private partnerships, grants, and budget allocations earmarked for digital transformation initiatives. This can achieve long-term cost savings, while meeting community needs effectively [224].
- *Stakeholder collaboration:* Collaboration with technology vendors, academic institutions, research organisations, and other government agencies is essential for accessing the expertise and resources necessary for successful AI implementation [90]. Partnerships facilitate knowledge exchange, innovation, and the co-creation of AI solutions tailored to local government challenges and objectives. Collaborative initiatives can involve joint research projects, shared data repositories, and collaborative platforms for testing and scaling AI applications [208].
- *Monitoring and evaluation:* Implementing the monitoring and evaluation mechanisms is critical for assessing the impact and effectiveness of AI initiatives within local governments [29]. Monitoring involves tracking key performance indicators, such as operational efficiencies, cost savings, and citizen satisfaction metrics, to measure the success of AI deployments [33]. Regular evaluation allows local governments to identify areas for improvement, address challenges proactively, and optimize AI systems based on real-world outcomes and feedback.
- *Pilot projects:* Conducting pilot projects provides local governments with valuable insights into the feasibility, scalability, and potential impact of AI technologies in real-world settings [225]. Pilot initiatives allow for the iterative testing and refinement of AI solutions before full-scale implementation, mitigating risks and optimising resource allocation [226].
- *Data governance:* Establishing data governance frameworks is essential for ensuring the quality, integrity, and ethical use of data in AI-driven initiatives within local governments [215]. Effective data governance involves defining policies and procedures for data collection, storage, sharing, and access control. It also includes implementing data management practices that uphold privacy regulations, promote data security, and mitigate risks associated with data breaches or misuse [29,156].
- *Regulatory compliance:* Compliance requirements may include data protection regulations, stakeholder-specific guidelines, and ethical standards for AI development and deployment, etc [158,159]. Local governments must stay abreast of evolving regulatory frameworks and adapt AI strategies and practices accordingly, to mitigate legal risks and ensure adherence to compliance obligations [207,217]. By proactively addressing regulatory considerations, local governments can foster a regulatory environment conducive to responsible AI innovation, while safeguarding public trust and upholding legal standards.

Adopting these interconnected strategies, local governments can navigate the complexities of AI implementation successfully and harness its transformative potential for the benefit of their communities.

5.6. Limitations of the Study

It is imperative to recognise the constraints that could affect how the results are interpreted: (a) This study is purely based on a grey literature review, which often lacks the rigorous peer-review process of academic literature, leading to variations in quality and reliability. (b) The analysis results presented here could involve the unintended bias of the authors. (c) There is a possibility that there exist more than 262 real-world use cases; however, our research methodology may not have been able to locate them, potentially resulting in gaps or the introduction of biases into the data synthesis process. (d) This study encountered challenges in accessing Chinese cases, which required additional efforts for data collection. This entailed conducting a secondary search using Baidu for rectification

and translation purposes, a process which was labour-intensive. (e) It is important to acknowledge the shortcomings in the available case information—each piece of literature reviewed presented large differences across their formats and the level of detail presented, ranging from comprehensive descriptions to brief overviews, limiting the depth of analysis and understanding. As a result, the key characteristics of the cases were restricted to AI technology, local government name, country name, a comprehensive use description, and year of introduction. While these factors provided a foundational understanding of each case, the varying levels of detail across sources necessitated the careful interpretation and synthesis of findings to derive meaningful insights. (f) Today, in many organisations, people are using Generative AI—such as ChatGPT—to perform routine office tasks without policy guidance or oversight or ‘under the radar’. This is almost certainly true in local governments too, especially in citizen-facing roles, but will not show up in an online trawl, but it certainly represents an important local government use case. Nonetheless, despite these limitations, the study was carried out in a measured and transparent manner, aiming to provide valuable insights into AI adoption in local governance, while considering these gaps in the methodology and data sources.

6. Conclusions

In recent years, AI has become a crucial urban phenomenon, serving as both a technology and infrastructure that supports city and community growth and as a burgeoning industry. Select cities worldwide have emerged as hubs of AI innovation and production, driving advancements and integrating AI into various aspects of urban life (Yigitcanlar et al., 2024). Consequently, this study aims to develop a comprehensive understanding of AI technology utilisation in smart city local government service provision and to generate lessons and best practices for similar smart city initiatives.

This study’s novelty lies in its pioneering analysis and in-depth review, providing invaluable insights into real-world scenarios where AI is employed by local governments. Additionally, it underscores the growing significance of AI in smart city governance, highlighting its potential to transform public service delivery, reshape decision-making processes, and redefine interactions between governments and citizens, particularly in the context of advancing the smart city agenda.

While AI presents promising solutions for enhancing local government efficiency, improving public services, and tackling complex challenges, its expanding integration into local governance necessitates a comprehensive and appropriate strategy to ensure AI will be responsibly used within the local government framework to enhance residents’ and community wellbeing. Furthermore, local governments should carefully consider ethical implications, data privacy, and the equitable distribution of technology benefits, ensuring that AI serves as a public value tool and fosters more responsive, efficient, and inclusive local governance.

We advocate the critical need for interdisciplinary collaboration in advancing research, practice, and policy related to the utilisation of AI technologies in local government contexts. By combining expertise from various fields, such as computer science, public administration, policy analysis, urban planning, philosophy, and social sciences, interdisciplinary collaboration can enrich the research in this area.

In terms of research, interdisciplinary collaboration can facilitate the integration of multiple perspectives and methodologies, enriching the analysis of AI utilisation in local government. For example, collaboration between computer scientists and scholars in public administration can foster the development of innovative research methodologies for data collection and analysis. This includes the application of NLP techniques to derive insights from policy documents and organisational reports, showcasing an interdisciplinary approach’s potential to yield significant advancements in understanding and applying AI in public sector contexts. Similarly, a collaboration between urban planners and social scientists can offer crucial insights into the societal impacts of AI adoption in local government. This includes exploring key issues related to equity, accessibility, and

community engagement, thereby deepening understanding of the impact of AI on different communities and social groups and helping to develop more inclusive and equitable public service strategies.

In addition, prospective research should aim to address the identified constraints in the existing research efforts, such as conducting more thorough practice reviews, improving data collection methodologies, and exploring the impact of AI on governance outcomes. Interdisciplinary collaboration can help address these constraints. By drawing on expertise from various disciplines, researchers can enhance the rigor and validity of their studies, ensuring that findings are robust and reliable. For example, interdisciplinary research teams can employ mixed methods approaches that combine quantitative data analysis with qualitative insights from interviews, focus groups, and participant observations. Adopting this holistic approach allows for a more nuanced understanding of the complex dynamics underlying AI utilisation in local government, which may help address multifaceted challenges and leverage the opportunities presented by AI technologies to enhance local governance and public service delivery.

Concerning practice, interdisciplinary collaboration can facilitate knowledge exchange and capacity building among practitioners from different fields. For example, collaboration between AI developers and public sector managers can promote the co-design and co-creation of AI-enabled solutions that are tailored to the specific needs and priorities of local government agencies. Similarly, a collaboration between data scientists and urban planners can support the development of data-driven decision-making frameworks that integrate AI technologies into the planning and implementation of urban development projects. Moreover, local government practitioners can benefit from the insights provided in this study by leveraging AI technologies to enhance service delivery, streamline administrative processes, and improve citizen engagement. Practitioners should also consider the ethical and societal implications of AI adoption and develop strategies to mitigate potential risks.

Regarding policy, interdisciplinary collaboration can inform the development of evidence-based policies and regulatory frameworks that govern the responsible use of AI in local government. By bringing together policymakers, legal experts, ethicists, and technology specialists, interdisciplinary collaboration can facilitate discussions on key policy issues, such as data privacy and security, algorithmic bias, fairness, transparency, and accountability. This collaborative approach can help ensure that AI policies are informed by the latest research findings and reflect the diverse perspectives of stakeholders from different disciplines. This includes establishing frameworks for ethical AI deployment, ensuring transparency and accountability in decision-making processes, and promoting equity and inclusivity in the access to AI-enabled services.

Overall, interdisciplinary collaboration is essential for advancing the research, practice, and policy related to AI utilisation in local government contexts. By leveraging the collective expertise and perspectives of diverse disciplines, interdisciplinary collaboration can drive innovation, foster knowledge exchange, and promote responsible and inclusive AI governance.

In conclusion, this study provides useful insights and perspectives for smart city local government decision-makers, practitioners, researchers, and other stakeholders to effectively utilise AI technology in the local government context. By enhancing the understanding of AI technology utilisation in local governments through the lessons drawn from 262 leading practices, this study lays a foundation for informed decision-making and strategic planning in local governance—particularly in the context of smart cities.

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Appendix A. Local Government AI Use Cases

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Inductive logic programming"	California cities	US	Transportation and urban planning	Building regulations	2019	" http://logicprogramming.stanford.edu/readings/symbium.pdf (accessed on 8 January 2024)"
"Robotic process automation"	Sea Girt	US	Transportation and urban planning	Building regulations	2018	" https://www.govpilot.com/blog/robotic-process-automation-for-local-governments/ (accessed on 19 January 2024)"
"Robotic process automation"	Norfolk County Council	UK	Administrative services	Local tax collection	2021	" https://www.blueprism.com/resources/case-studies/norfolk-county-council-enhances-citizens-experience-with-a-digital-workforce/ (accessed on 21 January 2024)"
"Robotic process automation"	Brent Council	UK	Transportation and urban planning	Housing services	2018	" https://www.uipath.com/resources/automation-case-studies/brent-council-uk-government-rpa (accessed on 8 January 2024)"
"Robotic process automation"	Surrey County Council	UK	Administrative services	Back-office work	2018	" https://www.uipath.com/resources/automation-case-studies/surrey-county-council-improves-employee-experience-with-automation (accessed on 10 January 2024)"
"Robotic process automation"	Municipality of Strängnäs	Sweden	Administrative services	Back-office work	2019	" https://www.uipath.com/resources/automation-case-studies/strangnas-municipality-government-rpa (accessed on 12 January 2024)"
"Robotic process automation"	Municipality of Copenhagen	Denmark	Administrative services	Back-office work	2015	" https://www.uipath.com/resources/automation-case-studies/copenhagen-municipality-enterprise-rpa#:~:text=Copenhagen%20has%20deployed%20its%20first,the%20information%20retrieval%20and%20reconciliation (accessed on 12 January 2024)"
"Robotic process automation"	Sefton Council	UK	Administrative services	Local tax collection	2015	" https://www.arvato.co.uk/wp-content/uploads/2019/06/Arvato_UK_rpa_public_sector_whitepaper_updated.pdf (accessed on 21 January 2024)"
"Robotic process automation"	Sefton Council	UK	Administrative services	Back-office work	2015	
"Robotic process automation"	North Tyneside Council	UK	Transportation and urban planning	Housing services	2017	" https://www.ukauthority.com/articles/robots-deliver-award-winning-customer-service-in-north-tyneside/ (accessed on 24 January 2024)"
"Robotic process automation"	North Tyneside Council	UK	Administrative services	Local tax collection	2017	
"Robotic process automation"	Cumbria County Council	UK	Transportation and urban planning	Transportation and traffic management	2022	" https://www.ukauthority.com/articles/automation-as-a-weapon-in-local-government-s-new-battles/ (accessed on 8 January 2024)"
"Robotic process automation"	Cumbria County Council	UK	Administrative services	Back-office work	2022	
"Robotic process automation"	Willoughby Council	Australia	Administrative services	Back-office work	2021	" https://zinno.com/automation/intelligent-automation-driving-government-digital-transformation-blog/ (accessed on 9 January 2024)"
"Robotic process automation"	Willoughby Council	Australia	Administrative services	Back-office work	2020	" https://www.governmentnews.com.au/type_contributors/dexter-the-robot-improving-customer-experience/ (accessed on 23 January 2024)"
"Robotic process automation"	San Francisco Municipality	San Francisco	Transportation and urban planning	Transportation and traffic management	2023	" https://zinno.com/automation/intelligent-automation-driving-government-digital-transformation-blog/ (accessed on 15 January 2024)"
"Robotic process automation"	City Council of Geneva	Switzerland	Healthcare and wellbeing	Financial assistance and economic development	2021	" https://zinno.com/automation/intelligent-automation-driving-government-digital-transformation-blog/#:~:text=Further,%20the%20City%20Council%20of,audit%20and%20risk%20management%20processes (accessed on 16 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Robotic process automation"	Liverpool City Council	Australia	Administrative services	Back-office work	2023	" https://zinno.com/automation/intelligent-automation-driving-government-digital-transformation-blog/#:~:text=Further,%20the%20City%20Council%20of,audit%20and%20risk%20management%20processes (accessed on 15 January 2024)"
"Robotic process automation"	Tasman Sea Hawke's Bay Regional Council	New Zealand	Transportation and urban planning	Resident registry	2020	" https://zinno.com/automation/intelligent-automation-driving-government-digital-transformation-blog/ (accessed on 10 January 2024)"
"Robotic process automation"	Municipality of Frederiksberg	Denmark	Transportation and urban planning	Resident registry	2020	" https://www.fujitsu.com/global/images/gig5/CS_2020Aug_Frederiksberg-Municipality.pdf (accessed on 17 January 2024)"
"Robotic process automation"	Municipality of Frederiksberg	Denmark	Transportation and urban planning	Resident registry	2023	" https://www.fujitsu.com/global/images/gig5/CS_2020Aug_Frederiksberg-Municipality.pdf (accessed on 19 January 2024)"
"Robotic process automation"	Pecos	USA	Administrative services	Back-office work	2022	" https://govlaunch.com/collections/automation (accessed on 16 January 2024)"
"Robotic process automation"	Avondale	USA	Administrative services	Back-office work	2020	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Middlesbrough Council	England	Administrative services	Local tax collection	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Burnaby	Canada	Administrative services	Back-office work	2022	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Nottingham City Council	England	Administrative services	Back-office work	2022	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Leeds City Council	England	Administrative services	Local tax collection	2016	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Glenelg	Australia	Administrative services	Back-office work	2020	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Kingston	Australia	Environmental management	Waste collection and management	2022	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Grand Forks	USA	Transportation and urban planning	Transportation and traffic management	2018	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Thurrock Council	England	Healthcare and wellbeing	Financial assistance and economic development	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Gloucestershire County Council	England	Administrative services	Back-office work	2022	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Porto Alegre	Brazil	Transportation and urban planning	Permits granting and licensing	2022	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Sundsvall	Sweden	Administrative services	Information management	2022	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Alcorcón City Council	Spain	Environmental management	Waste collection and management	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Norwich City Council	England	Healthcare and wellbeing	Financial assistance and economic development	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Hamilton City Council	Canada	Environmental management	Waste collection and management	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Robotic process automation"	Ronneby	Sweden	Administrative services	Back-office work	2021	" https://govlaunch.com/collections/automation (accessed on 10 January 2024)"
"Robotic process automation"	Culiacán	Mexico	Transportation and urban planning	Planning application processing	2021	" https://govlaunch.com/collections/automation (accessed on 10 January 2024)"
"Robotic process automation"	Tuscaloosa Municipality	USA	Environmental management	Maintaining public amenities	2021	" https://govlaunch.com/collections/automation (accessed on 10 January 2024)"
"Robotic process automation"	Värmdö	Sweden	Environmental management	Water and sewerage services	2023	" https://govlaunch.com/collections/automation (accessed on 10 January 2024)"
"Robotic process automation"	Sunshine Coast	Australia	Environmental management	Waste collection and management	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Devonport	Australia	Transportation and urban planning	Permits granting and licensing	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Heinola	Finland	Administrative services	Back-office work	2021	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Bellevue	USA	Transportation and urban planning	Permits granting and licensing	2020	" https://govlaunch.com/collections/automation (accessed on 8 January 2024)"
"Robotic process automation"	Tandridge District Council	England	Administrative services	Back-office work	2017	" https://govlaunch.com/collections/automation (accessed on 10 January 2024)"
"Robotic process automation"	Blacktown	Australia	Environmental management	Water and sewerage services	2020	" https://govlaunch.com/collections/automation (accessed on 10 January 2024)"
"Robotic process automation"	Auckland	New Zealand	Administrative services	Local tax collection	2020	" https://govlaunch.com/collections/automation (accessed on 10 January 2024)"
"Robotic process automation"	South Ayrshire Council	Scotland	Administrative services	Back-office work	2020	" https://www.theguardian.com/society/2020/oct/28/nearly-half-of-councils-in-great-britain-use-algorithms-to-help-make-claims-decisions (accessed on 27 January 2024)"
"Computer vision"	Erin	Canada	Environmental management	Maintaining public amenities	2021	" https://govlaunch.com/collections/automation (accessed on 15 January 2024)"
"Computer vision"	Stratford	Australia	Environmental management	Maintaining public amenities	2021	" https://govlaunch.com/collections/automation (accessed on 15 January 2024)"
"Computer vision"	Brimbank City Council	Australia	Environmental management	Waste collection and management	2023	" https://apo.org.au/sites/default/files/resource-files/2023-08/apo-nid323811_0.pdf (accessed on 22 January 2024)"
"Computer vision"	Kitchener	Canada	Environmental management	Maintaining public amenities	2021	" https://www.kitchener.ca/en/news/locally-made-robots-helping-city-staff-improve-kitchener-sidewalks.aspx (accessed on 23 January 2024)"
"Computer vision"	Municipality of Rotterdam	Netherlands	Transportation and urban planning	Building regulations	2022	" https://www.spotr.ai/customer-stories/rotterdam (accessed on 10 January 2024)"
"Computer vision"	City Council of A Western Australian	Australia	Healthcare and wellbeing	Leisure and recreation	2021	" https://www.integrasources.com/cases/computer-vision-sports-monitoring/ (accessed on 9 January 2024)"
"Computer vision"	Helsingborg Municipality	Sweden	Environmental management	Waste collection and management	2021	" https://univrses.com/press-releases/computer-vision-helps-make-helsingborg-a-smarter-city/ (accessed on 19 January 2024)"
"Computer vision"	Helsingborg Municipality	Sweden	Transportation and urban planning	Local road maintenance	2021	" https://univrses.com/press-releases/computer-vision-helps-make-helsingborg-a-smarter-city/ (accessed on 30 January 2024)"
"Computer vision"	Helsingborg Municipality	Sweden	Transportation and urban planning	Transportation and traffic management	2021	" https://univrses.com/press-releases/computer-vision-helps-make-helsingborg-a-smarter-city/ (accessed on 9 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
“Computer vision”	Tuscaloosa Municipality	USA	Environmental management	Waste collection and management	2021	“ https://www.planning.org/publications/report/9270237/ (accessed on 12 January 2024)”
“Computer vision”	Seoul	South Korea	Transportation and urban planning	Transportation and traffic management	2019	“ https://www.sparkcognition.com/artificial-intelligence-and-the-new-urban-infrastructure/ (accessed on 8 January 2024)”
“Computer vision”	Tel-Aviv Municipality	Israel	Environmental management	Water and sewerage services	2019	“ https://www.spiceworks.com/tech/iot/articles/what-is-internet-of-everthing/ (accessed on 10 January 2024)”
“Computer vision”	Las Vegas	USA	Transportation and urban planning	Transportation and traffic management	2021	“ https://governmenttechnologyinsider.com/whats-ahead-for-smart-cities/ (accessed on 9 January 2024)”
“Computer vision”	Mangaung Metropolitan Municipality	South Africa	Environmental management	Water and sewerage services	2019	“ https://www.smec.com/au/insights/deploying-artificial-intelligence-for-underground-asset-condition-assessments/ (accessed on 9 January 2024)”
“Computer vision”	Copenhagen City	Denmark	Environmental management	Local environmental issues	2019	“ https://www.linkedin.com/pulse/smart-cities-computer-vision-technology-debiprasad-bandopadhyay/ (accessed on 9 January 2024)”
“Computer vision”	Seoul	South Korea	Public safety and law enforcement	Public safety and security	2019	“ https://www.linkedin.com/pulse/smart-cities-computer-vision-technology-debiprasad-bandopadhyay/ (accessed on 22 January 2024)”
“Computer vision”	Singapore	Singapore	Transportation and urban planning	Transportation and traffic management	2018	“ https://www.linkedin.com/pulse/smart-cities-computer-vision-technology-debiprasad-bandopadhyay/ (accessed on 16 January 2024)”
“Computer vision”	Barcelona	Spain	Environmental management	Waste collection and management	2021	“ https://www.wowza.com/blog/smart-city-trends (accessed on 8 January 2024)”
“Computer vision”	Blackpool Council	England	Transportation and urban planning	Local road maintenance	2020	“ https://www.government-transformation.com/data/local-authorities-achieving-results-with-ai-roll-outs (accessed on 15 January 2024)”
“Computer vision”	BCP Council of Bournemouth, Christchurch, and Poole	England	Environmental management	Waste collection and management	2021	“ https://www.government-transformation.com/data/local-authorities-achieving-results-with-ai-roll-outs (accessed on 10 January 2024)”
“Natural language processing”	Milton Keynes	England	Transportation and urban planning	Permits granting and licensing	2018	“ https://www.government-transformation.com/data/local-authorities-achieving-results-with-ai-roll-outs (accessed on 19 January 2024)”
“Natural language processing”	Barcelona	Spain	Administrative services	Back-office work	2004	“ http://www.comune.torino.it/hops/documents/deliverables/brochure_A4_n1.pdf (accessed on 27 January 2024)”
“Natural language processing”	Turin Municipality	Italy	Administrative services	Back-office work	2004	“ http://www.comune.torino.it/hops/documents/deliverables/brochure_A4_n1.pdf (accessed on 15 January 2024)”
“Natural language processing”	London Borough of Camden	England	Administrative services	Back-office work	2004	“ http://www.comune.torino.it/hops/documents/deliverables/brochure_A4_n1.pdf (accessed on 10 January 2024)”
“Natural language processing”	Beirut Municipality	Lebanon	Administrative services	Community services—interpretation	2010	“ https://medium.com/beirut-spring/beirut-municipality-website-uses-machine-translation-to-populate-english-and-french-pages-590ff54b502c (accessed on 6 January 2024)”
“Natural language processing”	Wollongong City Council	Australia	Administrative services	Community services—interpretation	2022	“ https://wollongong.nsw.gov.au/about-google-translate (accessed on 6 January 2024)”

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Natural language processing"	Swindon Council	England	Administrative services	Community services—interpretation	2019	" https://cities-today.com/council-slashes-translation-costs-with-machine-learning/ (accessed on 6 January 2024)"
"Natural language processing"	Municipality of Rimini	Italy	Administrative services	Back-office work	2021	" https://dt4regions.eu/dt-book/dt-stories/open-digital-assistant (accessed on 8 January 2024)"
"Natural language processing"	Phoenix Municipality	USA	Administrative services	Community services—interpretation	2021	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 10 January 2024)"
"Natural language processing"	Williamsburg	USA	Administrative services	Information management	2018	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"
"Natural language processing"	Singapore	Singapore	Administrative services	Community services—complaints	2014	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"
"Natural language processing"	Kawasaki	Japan	Transportation and urban planning	Permits granting and licensing	2018	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"
"Natural language processing"	Kawasaki	Japan	Administrative services	Information management	2018	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"
"Natural language processing"	Kakegawa City	Japan	Transportation and urban planning	Permits granting and licensing	2018	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"
"Natural language processing"	Kolkata	India	Healthcare and wellbeing	Public health	2021	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 18 January 2024)"
"Natural language processing"	Kakegawa City	Japan	Administrative services	Information management	2018	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 18 January 2024)"
"Natural language processing"	Boston	USA	Healthcare and wellbeing	Public health	2021	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 18 January 2024)"
"Natural language processing"	Derby City Council	England	Healthcare and wellbeing	Public health	2023	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 18 January 2024)"
"Natural language processing"	Cabarrus County	USA	Healthcare and wellbeing	Public health	2021	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"
"Natural language processing"	Los Angeles	USA	Administrative services	Back-office work	2017	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Natural language processing"	Ronneby	Sweden	Administrative services	Back-office work	2021	" https://govlaunch.com/stories/ten-local-government-chatbots-that-are-making-a-difference (accessed on 15 January 2024)"
"Natural language processing"	San Antonio	USA	Healthcare and wellbeing	Leisure and recreation	2023	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Fairfield	USA	Administrative services	Community services—complaints	2017	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Derby City Council	England	Administrative services	Back-office work	2023	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Stirling	Scotland	Healthcare and wellbeing	Leisure and recreation	2023	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Coral Gables	USA	Administrative services	Information management	2023	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Atlanta	USA	Administrative services	Information management	2023	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Virginia Beach	USA	Administrative services	Back-office work	2023	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Nottingham City Council	England	Environmental management	Local environmental issues	2023	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Kelowna	Canada	Administrative services	Information management	2022	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Buenos Aires	Argentina	Healthcare and wellbeing	Financial assistance and economic development	2020	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Hamilton City Council	Canada	Administrative services	Information management	2022	" https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Kawasaki	Japan	Administrative services	Information management	2018	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Kawasaki	Japan	Transportation and urban planning	Permits granting and licensing	2018	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Phoenix Municipality	USA	Administrative services	Community services—Interpretation	2021	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Singapore	Singapore	Administrative services	Information management	2021	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Telford And Wrekin Council	England	Healthcare and wellbeing	Library maintenance	2021	" https://govlaunch.com/projects/telford-and-wrekin-council-add-three-new-services-thanks-to-tom-their-ai-assistant (accessed on 27 January 2024)"
"Natural language processing"	Telford and Wrekin Council	England	Transportation and urban planning	Housing services	2021	" https://govlaunch.com/projects/telford-and-wrekin-council-add-three-new-services-thanks-to-tom-their-ai-assistant (accessed on 30 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Natural language processing"	Telford and Wrekin Council	England	Transportation and urban planning	Resident registry	2021	" https://govlaunch.com/projects/telford-and-wrekin-council-add-three-new-services-thanks-to-tom-their-ai-assistant (accessed on 15 January 2024)"
"Natural language processing"	Lewes and Eastbourne Council	England	Administrative services	Back-office work	2022	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Frankston	Australia	Administrative services	Information management	2020	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Leeds City Council	England	Environmental management	Waste collection and management	2022	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Monmouthshire County Council	England	Administrative services	Information management	2021	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Grosseto	Italy	Administrative services	Information management	2021	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Treviso	Italy	Administrative services	Information management	2022	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Ronneby	Sweden	Administrative services	Back-office work	2021	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Mendoza	Argentina	Administrative services	Information management	2021	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Kortrijk	Belgium	Healthcare and wellbeing	Leisure and recreation	2021	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Boston	USA	Healthcare and wellbeing	Public health	2021	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Devonport	Australia	Administrative services	Information management	2021	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Houston	USA	Administrative services	Community services—complaints	2021	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	London Borough of Redbridge	London	Transportation and urban planning	Planning application processing	2020	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Kuusamo	Finland	Administrative services	Information management	2021	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Markham	Canada	Administrative services	Information management	2020	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Buenos Aires	Argentina	Transportation and urban planning	Transportation and traffic management	2021	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Markham	Canada	Administrative services	Information management	2020	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Buenos Aires	Argentina	Healthcare and wellbeing	Public health	2021	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	New Orleans	USA	Administrative services	Information management	2019	" https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Natural language processing"	Mogi Das Cruzes	Brazil	Transportation and urban planning	Resident registry	2021	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Sydney	Australia	Administrative services	Information management	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Trollhättan	Sweden	Administrative services	Community services—complaints	2023	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Delta	Canada	Healthcare and wellbeing	Public health	2021	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Manningham	Australia	Administrative services	Information management	2021	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Sønderborg	Denmark	Healthcare and wellbeing	Public health	2020	"https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Vantaa	Finland	Healthcare and wellbeing	Public health	2021	"https://govlaunch.com/collections/chatbots (accessed on 19 January 2024)"
"Natural language processing"	Duisburg	Germany	Healthcare and wellbeing	Burial grounds and electric crematorium	2020	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Varberg	Sweden	Administrative services	Information management	2021	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Hamilton City Council	New Zealand	Administrative services	Community feedback	2021	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Järvenpää	Finland	Administrative services	Information management	2020	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Tilburg	Netherlands	Administrative services	Information management	2022	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Porvoo	Finland	Administrative services	Information management	2022	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Borås	Sweden	Administrative services	Information management	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Pori	Finland	Administrative services	Back-office work	2021	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Greater Sudbury	Canada	Administrative services	Community services—complaints	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Knoxville	USA	Administrative services	Community services—interpretation	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Kelowna	Canada	Healthcare and wellbeing	Public health	2020	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Maribyrnong	Australia	Administrative services	Community services—interpretation	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Dallas	USA	Healthcare and wellbeing	Public health	2020	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Natural language processing"	Adelaide	Australia	Environmental management	Waste collection and management	2020	"https://govlaunch.com/collections/chatbots (accessed on 26 January 2024)"
"Natural language processing"	Adelaide	Australia	Transportation and urban planning	Transportation and traffic management	2020	"https://govlaunch.com/collections/chatbots (accessed on 26 January 2024)"
"Natural language processing"	Adelaide	Australia	Healthcare and wellbeing	Library maintenance	2020	"https://govlaunch.com/collections/chatbots (accessed on 26 January 2024)"
"Natural language processing"	Bellevue	USA	Healthcare and wellbeing	Public health	2020	"https://govlaunch.com/collections/chatbots (accessed on 26 January 2024)"
"Natural language processing"	Goldsboro	USA	Administrative services	Community services—complaints	2021	"https://govlaunch.com/collections/chatbots (accessed on 26 January 2024)"
"Natural language processing"	Portland	USA	Administrative services	Back-office work	2020	"https://govlaunch.com/collections/chatbots (accessed on 26 January 2024)"
"Natural language processing"	Arun District Council	England	Administrative services	Information management	2021	"https://govlaunch.com/collections/chatbots (accessed on 27 January 2024)"
"Natural language processing"	Derby City Council	England	Healthcare and wellbeing	Public health	2021	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Aberdeen City Council	Scotland	Administrative services	Information management	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Williamsburg	USA	Administrative services	Back-office work	2018	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Johns Creek	USA	Administrative services	Information management	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Ottawa	Canada	Environmental management	Waste collection and management	2019	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Austin	USA	Healthcare and wellbeing	Public health	2020	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Johns Creek	USA	Administrative services	Back-office work	2018	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Los Angeles	USA	Administrative services	Back-office work	2017	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	North Charleston	USA	Administrative services	Community services—complaints	2018	"https://govlaunch.com/collections/chatbots (accessed on 18 January 2024)"
"Natural language processing"	Kansas City	USA	Administrative services	Information management	2017	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Henderson	USA	Administrative services	Information management	2019	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Johns Creek	USA	Administrative services	Information management	2018	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Virginia Beach	USA	Administrative services	Information management	2021	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Albuquerque	USA	Administrative services	Information management	2017	"https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Natural language processing"	Williamsburg	USA	Administrative services	Information management	2018	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	Gilbert	USA	Administrative services	Community feedback	2018	" https://govlaunch.com/collections/chatbots (accessed on 15 January 2024)"
"Natural language processing"	San Jose	USA	Administrative services	Community services—complaints	2020	" https://www.govtech.com/opinion/how-ai-helps-state-and-local-governments-work-smarter (accessed on 18 January 2024)"
"Neural Network"	Chicago's Local Government	USA	Public safety and law enforcement	Public safety and security	2018	" https://d3.harvard.edu/platform-rctom/submission/smarter-cities-how-machine-learning-can-improve-municipal-services-in-chicago/ (accessed on 27 January 2024)"
"Neural Network"	Cartagena, Medellin, and Monteria	Colombia	Administrative services	Community feedback	2020	" https://www.oecd-ilibrary.org/sites/08955f48-en/index.html?itemId=/content/component/08955f48-en (accessed on 19 January 2024)"
"Neural Network"	Los Angeles	USA	Transportation and urban planning	Housing services	2018	" https://www.govtech.com/opinion/how-ai-helps-state-and-local-governments-work-smarter (accessed on 16 January 2024)"
"Neural Network"	North Tyneside Council	England	Administrative services	Local tax collection	2021	" https://www.theguardian.com/society/2019/oct/15/councils-using-algorithms-make-welfare-decisions-benefits (accessed on 19 January 2024)"
"Neural Network"	Hackney Council	England	Public safety and law enforcement	Public safety and security	2021	" https://www.theguardian.com/society/2019/oct/15/councils-using-algorithms-make-welfare-decisions-benefits (accessed on 19 January 2024)"
"Neural Network"	Municipality of Amsterdam	Netherlands	Transportation and urban planning	Transportation and traffic management	2022	" https://www.xomnia.com/xomnia-supports-the-municipality-of-amsterdam-with-machine-learning-expertise/ (accessed on 19 January 2024)"
"Neural Network"	City of Ryde	Australia	Environmental management	Urban forestry	2020	" https://www.govtechreview.com.au/content/gov-tech-review-topics/white-paper/local-government-stays-green-with-machine-learning-783314431/preparing_download (accessed on 17 January 2024)"
"Neural Network"	Swindon Borough Council	England	Administrative services	Community services—interpretation	2021	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	Buffalo	USA	Environmental management	Water and sewerage services	2023	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	Irving	USA	Transportation and urban planning	Transportation and traffic management	2023	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	East Lansing	USA	Environmental management	Waste collection and management	2022	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	Lancashire County Council	England	Transportation and urban planning	Transportation and traffic management	2022	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	North Tyneside Council	England	Healthcare and wellbeing	Public health	2022	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	Aberdeen City Council	Scotland	Healthcare and wellbeing	Public health	2022	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	Gilbert	USA	Public safety and law enforcement	Public safety and security	2019	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Neural Network"	Sunderland City Council	England	Public safety and law enforcement	Public safety and security	2022	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	Sunderland City Council	England	Administrative services	Local tax collection	2022	" https://govlaunch.com/collections/machine-learning (accessed on 22 January 2024)"
"Neural Network"	Philadelphia	USA	Environmental management	Maintaining public amenities	2021	" https://govlaunch.com/collections/machine-learning (accessed on 21 January 2024)"
"Neural Network"	Los Angeles	USA	Transportation and urban planning	Transportation and traffic management	2022	" https://ascend.thentia.com/process/applications-of-machine-learning-in-digital-government/ (accessed on 19 January 2024)"
"Neural Network"	City of Atlanta	USA	Transportation and urban planning	Transportation and traffic management	2017	" https://ascend.thentia.com/process/applications-of-machine-learning-in-digital-government/ (accessed on 17 January 2024)"
"Neural Network"	Kansas City	USA	Transportation and urban planning	Transportation and traffic management	2017	" https://ascend.thentia.com/process/applications-of-machine-learning-in-digital-government/ (accessed on 29 January 2024)"
"Autonomous System"	Ogaki City	Japan	Administrative services	Information management	2020	" https://www.japantimes.co.jp/news/2019/01/15/national/city-hall-gifu-prefecture-first-japan-deploy-autonomous-robots-aid-residents/ (accessed on 16 January 2024)"
"Autonomous System"	Pittsburgh	USA	Environmental management	Water and sewerage services	2016	" https://www.automate.org/blogs/autonomous-robots-are-moving-from-below-the-streets-and-on-to-highways (accessed on 15 January 2024)"
"Autonomous System"	Upplands-Bro Municipality	Sweden	Healthcare and wellbeing	Public health	2020	" https://www.smartcitiesworld.net/news/swedish-municipality-deploys-robots-for-safer-recruitment-5251 (accessed on 9 January 2024)"
"Autonomous System"	Municipalities in Finland	Finland	Healthcare and wellbeing	Public health	2016	" https://www.sciencedirect.com/science/article/pii/S1386505619300498?ref=pdf_download&df=RR-2&rr=8381b903ac20a7ff (accessed on 15 January 2024)"
"Autonomous System"	Pune Municipality	India	Environmental management	Maintaining public amenities	2022	" https://ilougemia.com/pune-municipal-corporation-introduces-advanced-robots-to-clean-manholes/ (accessed on 17 January 2024)"
"Autonomous System"	Bucher Municipality	Singapore	Transportation and urban planning	Local road maintenance	2020	" https://www.buchermunicipal.com/int/news/bucher-municipal-acquires-enway (accessed on 18 January 2024)"
"Autonomous System"	London Borough	England	Transportation and urban planning	Permits granting and licensing	2016	" https://www.theguardian.com/public-leaders-network/2016/jul/04/robot-amelia-future-local-government-enfield-council (accessed on 18 January 2024)"
"Autonomous System"	Ku-ring-gai Council	Australia	Public safety and law enforcement	Public safety and security	2019	" https://www.climatechange.environment.nsw.gov.au/sites/default/files/2022-09/Simtable_modelling_toolKu-ring-gai_Council.pdf (accessed on 18 January 2024)"
"Autonomous System"	Hangzhou	China	Transportation and urban planning	Transportation and traffic management	2019	" https://www.hangzhou.gov.cn/art/2021/12/24/art_812262_59046787.html?eqid=f360863400062d0b000000026486e33e (accessed on 22 January 2024)"
"Autonomous System"	Hangzhou	China	Transportation and urban planning	Town planning	2019	" https://www.hangzhou.gov.cn/art/2021/12/24/art_812262_59046787.html?eqid=f360863400062d0b000000026486e33e (accessed on 19 January 2024)"
"Autonomous System"	Hangzhou	China	Healthcare and wellbeing	Financial assistance and economic development	2019	" https://www.hangzhou.gov.cn/art/2021/12/24/art_812262_59046787.html?eqid=f360863400062d0b000000026486e33e (accessed on 15 January 2024)"
"Neural Network"	Hangzhou	China	Healthcare and wellbeing	Leisure and recreation	2019	" https://www.hangzhou.gov.cn/art/2021/12/24/art_812262_59046787.html?eqid=f360863400062d0b000000026486e33e (accessed on 16 January 2024)"

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
“Neural Network”	Hangzhou	China	Healthcare and wellbeing	Public health	2019	“ https://www.hangzhou.gov.cn/art/2021/12/24/art_812262_59046787.html?eqid=f36086340062d0b000000026486e33e (accessed on 27 January 2024)”
“Computer vision”	Hangzhou	China	Transportation and urban planning	Transportation and traffic management	2016	“ http://www.cac.gov.cn/2018-11/27/c_1123771419.htm?isappinstalled=0 (accessed on 3 January 2024)”
“Natural Language Processing”	Guiyang	China	Administrative services	Community services—complaints	2018	“ http://www.cac.gov.cn/2018-11/27/c_1123771419.htm?isappinstalled=0 (accessed on 3 January 2024)”
“Computer vision”	Shenzhen	China	Transportation and urban planning	Transportation and traffic management	2018	“ http://www.cac.gov.cn/2018-11/27/c_1123771419.htm?isappinstalled=0 (accessed on 3 January 2024)”
“Natural Language Processing”	Shanghai	China	Administrative services	Community services—interpretation	2018	“ https://www.sast.gov.cn/content.html?id=kjb228884 (accessed on 6 January 2024)”
“Computer vision”	Chengdu	China	Environmental management	River management	2018	“ https://www.sc.gov.cn/10462/10778/10876/2024/1/10/f30e99b8b89947b895a7399b114c3152.shtml (accessed on 8 January 2024)”
“Robotic process automation”	Yanan	China	Transportation and urban planning	Planning application processing	2018	“ http://www.cac.gov.cn/2018-06/03/c_1122925064.htm (accessed on 15 January 2024)”
“Computer vision”	Guangzhou	China	Transportation and urban planning	Permits granting and licensing	2019	“ http://www.cac.gov.cn/2019-10/25/c_1573534978283427.htm (accessed on 8 January 2024)”
“Computer vision”	Wuhan	China	Transportation and urban planning	Transportation and traffic management	2019	“ http://www.mod.gov.cn/gfbw/gfjy_index/zyhd/4852807.html (accessed on 9 January 2024)”
“Neural Network”	Changsha	China	Administrative services	Information management	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Neural Network”	Changsha	China	Environmental management	Waste collection and management	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Neural Network”	Changsha	China	Public safety and law enforcement	Public safety and security	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Natural Language Processing”	Changsha	China	Administrative services	Information management	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Neural Network”	Changsha	China	Administrative services	Information management	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Computer vision”	Changsha	China	Transportation and urban planning	Transportation and traffic management	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Computer vision”	Changsha	China	Environmental management	Waste collection and management	2021	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Neural Network”	Changsha	China	Healthcare and wellbeing	Financial assistance and economic development	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Neural Network”	Changsha	China	Administrative services	Information management	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 9 January 2024)”
“Neural Network”	Changsha	China	Public safety and law enforcement	meteorological services	2020	“ http://www.tianxin.gov.cn/zjtx23/ytx67/mtjj4/202006/t20200601_8156353.html (accessed on 23 January 2024)”

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
“Computer vision”	Changsha	China	Transportation and urban planning	Town planning	2020	“ http://www.tianxin.gov.cn/zjtx23/yt67/mtij4/202006/t20200601_8156353.html (accessed on 23 January 2024)”
“Neural Network”	Chongqing	China	Transportation and urban planning	Permits granting and licensing	2020	“ https://www.ndrc.gov.cn/xwdt/ztzl/szhzxhbxg/gfdt/202007/t20200713_1233659.html (accessed on 23 January 2024)”
“Neural Network”	Chongqing	China	Transportation and urban planning	Transportation and traffic management	2020	“ https://www.ndrc.gov.cn/xwdt/ztzl/szhzxhbxg/gfdt/202007/t20200713_1233659.html (accessed on 23 January 2024)”
“Computer vision”	Chongqing	China	Transportation and urban planning	Town planning	2020	“ https://www.ndrc.gov.cn/xwdt/ztzl/szhzxhbxg/gfdt/202007/t20200713_1233659.html (accessed on 23 January 2024)”
“Neural Network”	Chongqing	China	Environmental management	Local environmental issues	2020	“ https://www.ndrc.gov.cn/xwdt/ztzl/szhzxhbxg/gfdt/202007/t20200713_1233659.html (accessed on 23 January 2024)”
“Neural Network”	Chongqing	China	Transportation and urban planning	Town planning	2020	“ https://www.ndrc.gov.cn/xwdt/ztzl/szhzxhbxg/gfdt/202007/t20200713_1233659.html (accessed on 23 January 2024)”
“Natural Language Processing”	Huhehaote	China	Environmental management	Maintaining public amenities	2020	“ https://zwfw.nmg.gov.cn/pub/fwzx/202012/t20201224_19302.html (accessed on 23 January 2024)”
“Autonomous System”	Chongqing	China	Administrative services	Information management	2020	“ http://www.wz.gov.cn/zwxx_266/jdtp/202009/t20200917_7890266_wap.html (accessed on 23 January 2024)”
“Robotic process automation”	Hangzhou	China	Environmental management	Water and sewerage services	2021	“ http://www.linan.gov.cn/art/2021/10/19/art_1229601278_59061028.html (accessed on 26 January 2024)”
“Neural Network”	Hangzhou	China	Healthcare and wellbeing	Pest control services	2021	“ https://www.linan.gov.cn/art/2021/10/19/art_1229601278_59061028.html (accessed on 17 January 2024)”
Autonomous System	Anyang	China	Administrative services	Information management	2021	“ https://dsj.henan.gov.cn/2021/09-26/2318831.html (accessed on 19 January 2024)”
“Computer vision”	Guangzhou	China	Transportation and urban planning	Transportation and traffic management	2022	“ https://www.hp.gov.cn/xwzx/mbxx/content/post_8663139.html (accessed on 23 January 2024)”
“Computer vision”	Weihai	China	Administrative services	Information management	2022	“ http://www.wendeng.gov.cn/art/2022/9/8/art_99344_2970189.html (accessed on 16 January 2024)”
“Neural Network”	Beijing	China	Administrative services	Information management	2022	“ https://www.bjtz.gov.cn/bjtz/xxfb/202208/1610401.shtml (accessed on 15 January 2024)”
“Neural Network”	Beijing	China	Administrative services	Information management	2022	“ https://www.beijing.gov.cn/ywdt/gqrd/202203/t20220304_2622495.html (accessed on 19 January 2024)”
“Natural Language Processing”	Hanzhong	China	Administrative services	Information management	2023	“ http://www.hanzhong.gov.cn/hzszf/xwzx/bmdt/202307/aa783e1c4a8f4f2d9f9a6b3abb5f735f.shtml (accessed on 22 January 2024)”
“Natural Language Processing”	Yinchuan	China	Administrative services	Community services—complaints	2024	“ https://www.gov.cn/govweb/lianbo/difang/202401/content_6925551.htm (accessed on 22 January 2024)”
“Robotic process automation”	Jinan	China	Transportation and urban planning	Permits granting and licensing	2022	“ http://www.jinan.gov.cn/art/2022/8/22/art_80993_4926510.html (accessed on 19 January 2024)”

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
“Computer vision”	Harbin	China	Transportation and urban planning	Permits granting and licensing	2021	“ https://www.ndrc.gov.cn/fggz/fggz/dfxx/202109/t20210917_1296931.html (accessed on 17 January 2024)”
“Autonomous System”	Jiabin	China	Administrative services	Information management	2020	“ https://www.jiabin.gov.cn/art/2020/9/29/art_1685305_58831028.html (accessed on 10 January 2024)”
“Robotic process automation”	Shenzhen	China	Healthcare and wellbeing	Public health	2017	“ http://ka.sz.gov.cn/ztzl/zt001/content/post_2291748.html (accessed on 10 January 2024)”
“Neural Network”	Beijing	China	Transportation and urban planning	Transportation and traffic management	2017	“ https://jtgl.beijing.gov.cn/jgj/jgxx/94246/95332/537586/index.html (accessed on 13 January 2024)”
“Robotic process automation”	Shanghai	China	Environmental management	Maintaining public amenities	2018	“ https://www.shwm.gov.cn/TrueCMS/shwmw/xyesdxzzsh/content/2d025728-f2d0-4980-87c7-a04db74ce82b.html (accessed on 13 January 2024)”
“Robotic process automation”	Shanghai	China	Environmental management	River management	2018	“ https://www.shwm.gov.cn/TrueCMS/shwmw/xyesdxzzsh/content/2d025728-f2d0-4980-87c7-a04db74ce82b.html (accessed on 12 January 2024)”
“Robotic process automation”	Shenzhen	China	Healthcare and wellbeing	Public health	2019	“ http://wjw.sz.gov.cn/ztzl/ygsn/cxal/content/post_3119990.html (accessed on 23 January 2024)”
“Neural Network”	Shenzhen	China	Transportation and urban planning	Transportation and traffic management	2019	“ http://jtys.sz.gov.cn/zwgk/ztzl/mss/2019wcr/mrhd/content/post_4205133.html (accessed on 29 January 2024)”
“Robotic process automation”	Shenzhen	China	Transportation and urban planning	Transportation and traffic management	2020	“ http://www.szss.gov.cn/sstbhq/qtdy/zlqmdqyqfkhfgfcssl/content/post_7387911.html (accessed on 18 January 2024)”
“Autonomous System”	Shanghai	China	Transportation and urban planning	Transportation and traffic management	2021	“ http://jtyst.jiangsu.gov.cn/art/2021/2/5/art_41775_9666644.html (accessed on 16 January 2024)”
“Neural Network”	Shenzhen	China	Environmental management	Urban forestry	2022	“ http://meeb.sz.gov.cn/gkmlpt/content/10/10159/post_10159242.html#3765 (accessed on 18 January 2024)”
“Neural Network”	Shanghai	China	Healthcare and wellbeing	Public health	2022	“ https://www.shanghai.gov.cn/gwk/search/content/7aa19db8864a41a39d111039617a49a7?eqid=862b4aad0002618c00000003647d95a7 (accessed on 8 January 2024)”
“Robotic process automation”	Beijing	China	Healthcare and wellbeing	Public health	2022	“ https://www.beijing.gov.cn/gate/big5/www.beijing.gov.cn/ywdt/zwzt/dah/bxyw/202201/t20220120_2596203.html (accessed on 8 January 2024)”
“Natural Language Processing”	Chengdu	China	Administrative services	Community feedback	2022	“ https://cdswszw.gov.cn/tzgg/Detail.aspx?id=27136 (accessed on 8 January 2024)”
“Computer vision”	Hangzhou	China	Environmental management	Waste collection and management	2022	“ http://epb.hangzhou.gov.cn/art/2022/12/7/art_1692261_59025412.html (accessed on 8 January 2024)”
“Robotic process automation”	Shenzhen	China	Healthcare and wellbeing	Public health	2022	“ https://www.sz.gov.cn/cn/xxgk/zfxgj/gqdt/content/post_10001478.html (accessed on 8 January 2024)”
“Neural Network”	Guangzhou	China	Healthcare and wellbeing	Public health	2022	“ https://www.gz.gov.cn/zfjg/gzsybzb/bmdt/content/post_8701201.html (accessed on 8 January 2024)”
Computer vision	Chengdu	China	Transportation and urban planning	Transportation and traffic management	2023	“ https://www.mot.gov.cn/jiaotongyaowen/202303/t20230302_3767032.html (accessed on 10 January 2024)”

AI Technology	Local Government	Country	General Service	Sub-Service	Year	URL
"Neural Network"	Shanghai	China	Administrative services	Information management	2023	" https://app.sheitc.sh.gov.cn/gydt/691296.htm (accessed on 10 January 2024)"
"Computer vision"	Beijing	China	Public safety and law enforcement	Public safety and security	2023	" https://www.beijing.gov.cn/fuwu/bmfw/sy/jrts/202304/t20230414_3032789.html (accessed on 15 January 2024)"
"Computer vision"	Guangzhou	China	Environmental management	Local environmental issues	2023	" http://gxj.gz.gov.cn/zt/dlys/aljd/content/post_9384756.html (accessed on 15 January 2024)"

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